Fundamentals of CMOS RF PLL Frequency Synthesizer

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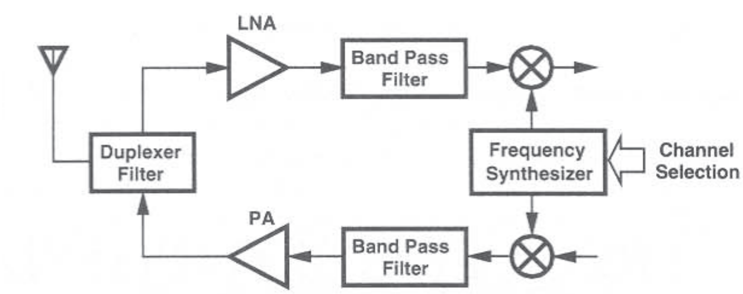
PLL Synthesizer in RF Transceiver

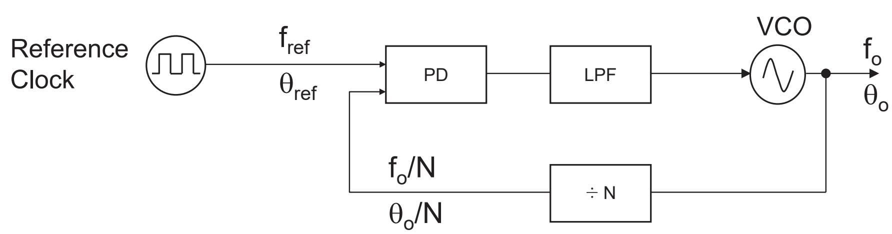


- Receiver:
  - LNA and Downconversion mixer
- Transmitter:
  - Upconversion mixer and power amplifier
- PLL:
  - Synthesizer provides LO signals for the frequency-converting mixers

Frequency Synthesizer

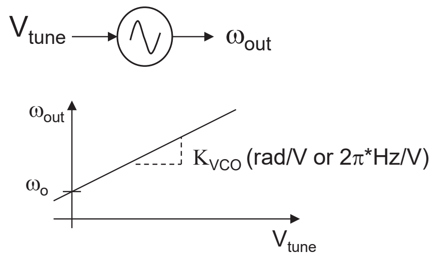
- Precise clock is always needed in wireless and wire-line communication systems
  - Phase locked loop (PLL) forces a free-running inaccurate oscillator to generate a precise stable clean wanted frequency
  - Frequency synthesizer synthesizes the output frequency as users want.
  - PLL synthesizer is a frequency synthesizer based on PLL
- Design Requirements
  - Output frequency accuracy
  - Frequency resolution
  - Spectral purity: Phase noise, Spur
  - Lock time
  - Power consumption, silicon area
- Architecture
  - PLL-based frequency synthesizer: Integer-N type, Fractional-N type

PLL-based Frequency Synthesizer Overview



- Operations:
  - Reference clock generates a clean and accurate, but low frequency (~ few MHz) signal f_{ref}
  - VCO generates an unstable and inaccurate, but very high frequency (~ few GHz) signal f_o
  - Divider ÷N divides the VCO signal f_o by N to produce f_o/N
  - Phase Detector (PD) converts phase error to corresponding voltage signal
  - LPF converts the PD output voltage to dc to control the VCO frequency
- Results:
  - Feedback loop works for locking the divided VCO signal to the reference clock → $f_{ref} = f_o/N$
  - Then, VCO output frequency can be tuned by adjusting the divide ratio N → Frequency synthesis: $f_o = N \times f_{ref}$
- Now, we need more study to understand the full operation principle

Voltage-Controlled Oscillator Transfer Function



$$\omega_{out} = \omega_o + K_{VCO} V_{tune}$$

$$\phi_{out}(t) = \int_{-\infty}^{t} (\omega_o + K_{VCO} V_{tune}) dt$$

$$V_{out}(t) = A\cos\left(\omega_o t + \underbrace{K_{VCO}\int_{-\infty}^{t} V_{tune} dt}_{\text{Excess phase}}\right)$$

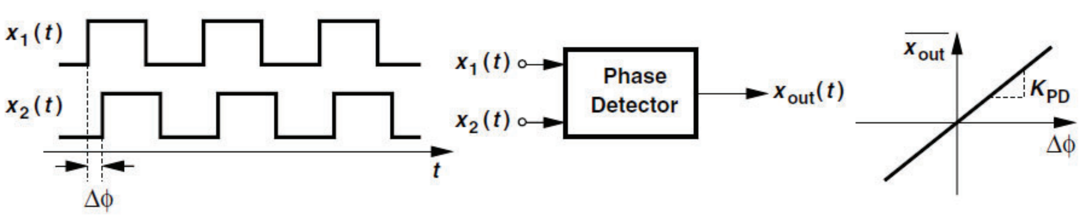
$$\phi_{vco}(t) = K_{VCO}\int V_{tune} dt$$

$$\frac{\phi_{vco}(s)}{V_{tune}(s)} = \frac{K_{VCO}}{s}$$

➔ VCO is an integrator of V_{tune}

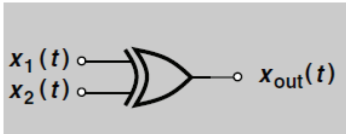
Phase Detector

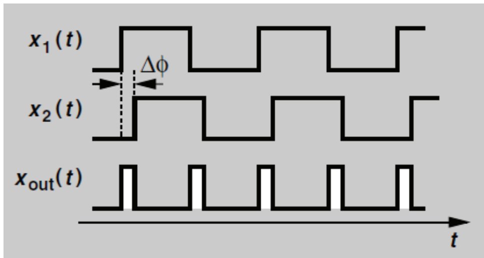
- Ideal PD produces dc output signal in proportion to the input phase difference.

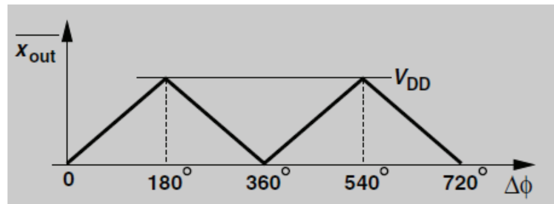


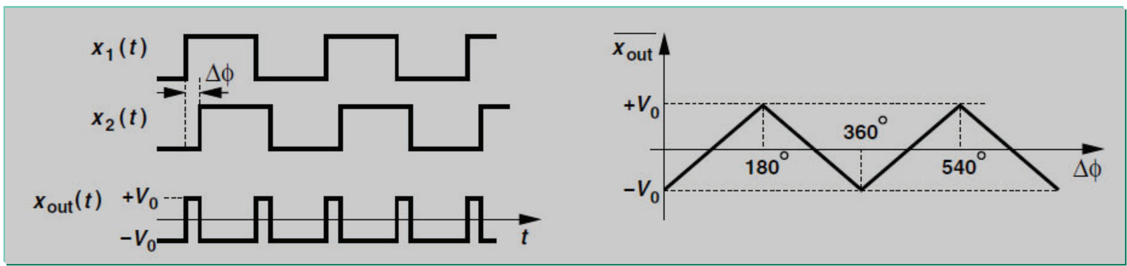
$$\overline{x_{out}} = K_{PD} \cdot \Delta\phi$$

Implementation of Phase Detector: XOR

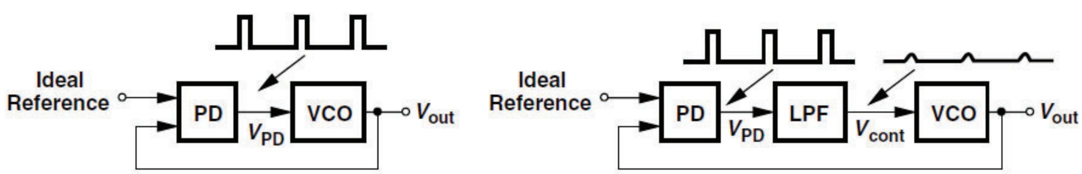


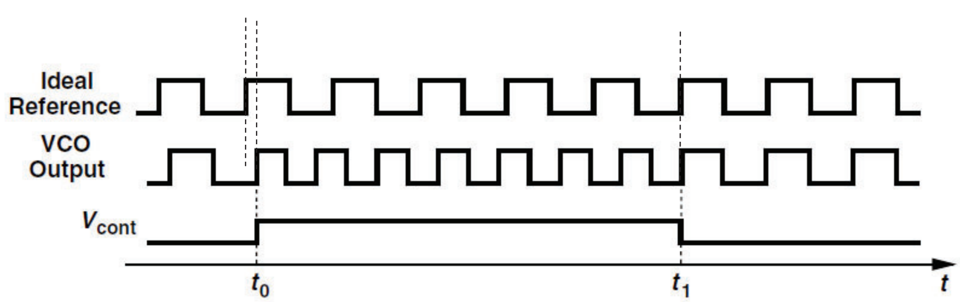






Alignment of VCO Phase to Reference Phase

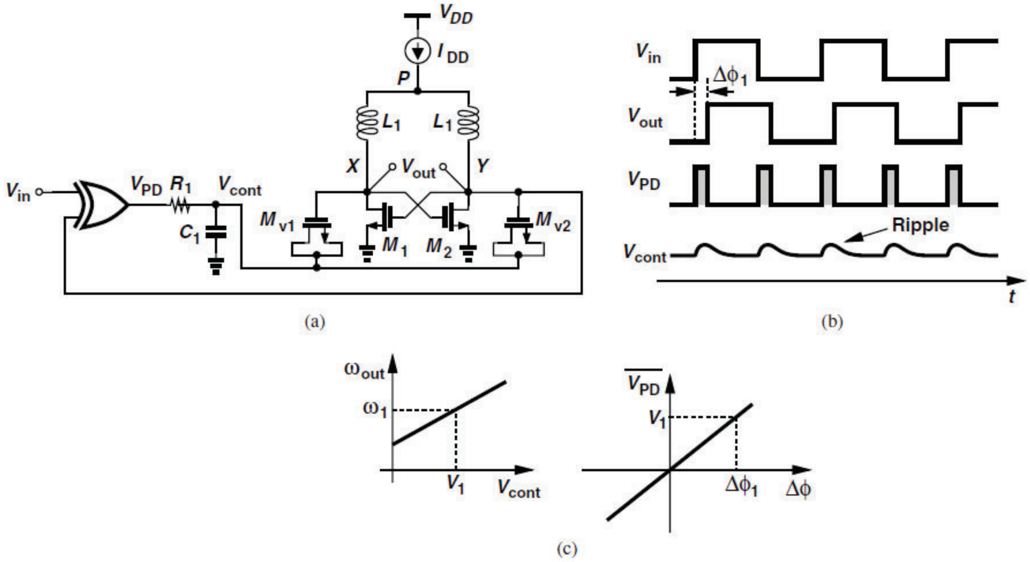




VCO frequency need to increase in order to achieve the phase alignment at t_1

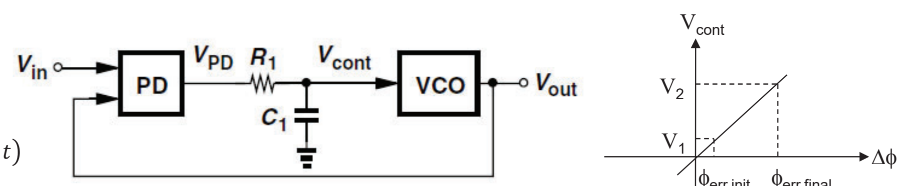
Note: It is not so simple in practice

A Simple PLL Implementation Example

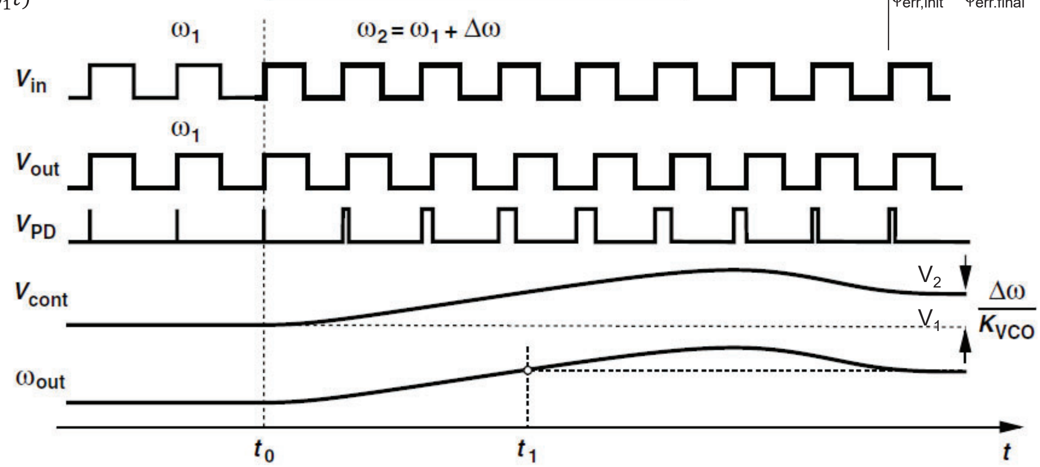


(a) PLL implementation example, (b) waveforms at different nodes, (c) VCO and PD input/output characteristics showing the system solution.

Frequency Step Response

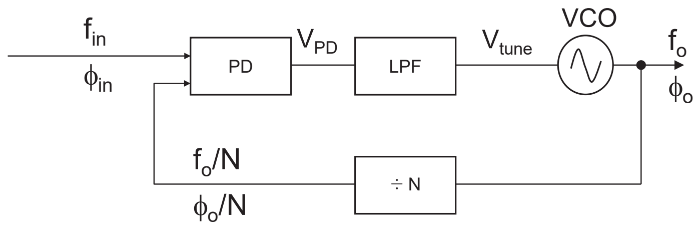


$V_{in}(t) = V_A \cos((\omega_1 + \Delta\omega \cdot u(t - t_1)) \cdot t)$
$V_{out}(t) = V_B \cos(\omega_1 t)$



Understanding Simple PLL Synthesizer Operation

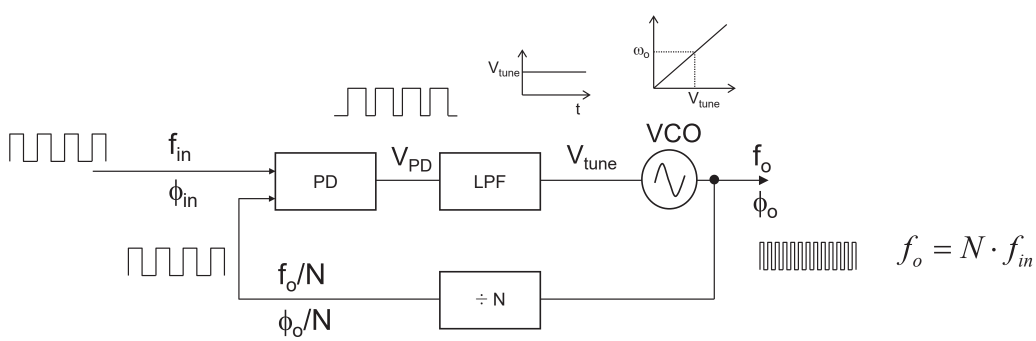
Simple PLL Synthesizer Architecture



- It is a negative feedback system with PD as an error amplifier
- The divided output phase (ϕ_o/N) is locked to the input phase (ϕ_{in})
  → Phase locking system (not frequency locking system, why ?)
- In "phase-locked" condition,

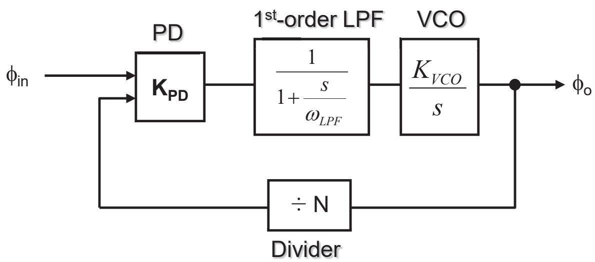
$$\varphi_{in} - \frac{\varphi_o}{N} = \text{constant} \Rightarrow \frac{d\left(\varphi_{in} - \frac{\varphi_o}{N}\right)}{dt} = 0 \quad \Rightarrow \quad f_{in} = \frac{f_o}{N} \Rightarrow \boxed{f_o = N \cdot f_{in}}$$

Waveforms in a Simple PLL Operation



※ also known as Type-I PLL

Linear Model of Type-I PLL



$$H(s)\Big|_{open} = \frac{\varphi_{out}}{\varphi_{in}}(s)\Big|_{open} = K_{PD} \cdot \frac{1}{1 + \frac{s}{\omega_{LPF}}} \cdot \frac{K_{VCO}}{s}$$

$$H(s)\Big|_{closed} = \frac{\varphi_{out}}{\varphi_{in}}(s) = N \cdot \frac{K_{PD}K'_{VCO}}{\frac{s^2}{\omega_{LPF}} + s + K_{PD}K'_{VCO}}$$

$$where\ \ K'_{VCO} = \frac{K_{VCO}}{N}$$

General Transfer Function of a Second-Order Low-Pass System

$$\Rightarrow H(s) = \frac{{\omega_n}^2}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$$

$*$ Natural Frequency: $\omega_n = \sqrt{\omega_{LPF}K_{PD}K'_{VCO}}$

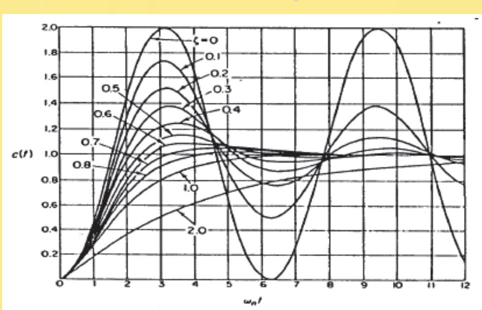
$*$ Damping Factor : $\zeta = \frac{1}{2}\sqrt{\frac{\omega_{LPF}}{K_{PD}K'_{VCO}}}$

Second-Order Feedback System Response

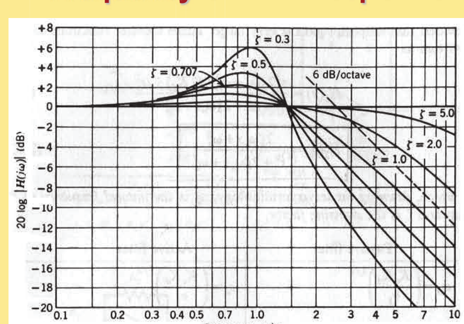
$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

where ω_n is natural frequency
ζ is damping factor

Time-domain response



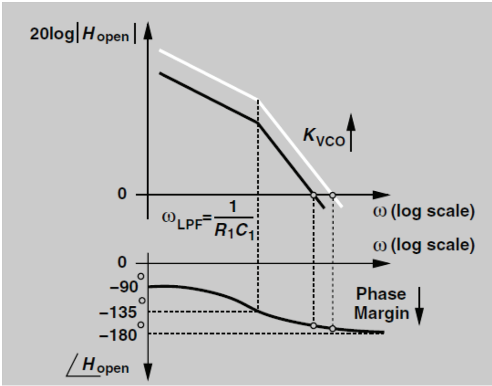
Frequency-domain response

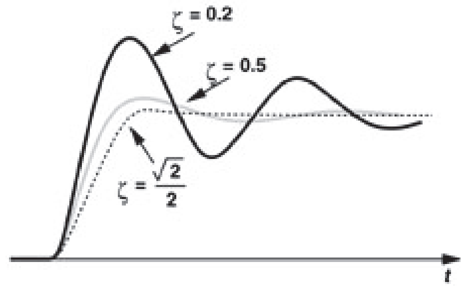


$$Optimal\ Design \rightarrow \zeta = \frac{1}{\sqrt{2}} \rightarrow K_{PD}K_{VCO} = \frac{\omega_{LPF}}{2}$$

Limitations of Simple PLL (1)

- Tight relation between the loop stability and K_{VCO}, ω_{LPF}





$$\zeta = \frac{1}{2}\sqrt{\frac{\omega_{LPF}}{K_{PD}K_{VCO}}}$$

$$\omega_n = \sqrt{\omega_{LPF}K_{PD}K_{VCO}}$$

$$\zeta\omega_n = \frac{1}{2}\omega_{LPF}$$

conflicting requirements

$$K_{VCO}\uparrow \Rightarrow PM \ or \ \zeta \downarrow$$

$$\omega_{LPF}\downarrow \Rightarrow PM \ or \ \zeta \downarrow$$

$$\text{Stability} \propto \zeta$$

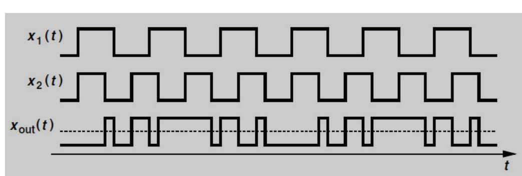
$$\text{Lock time} \propto \frac{1}{\omega_{LPF}}$$

$$\text{Phase Error} \propto \frac{1}{K_{PD}K_{VCO}}$$

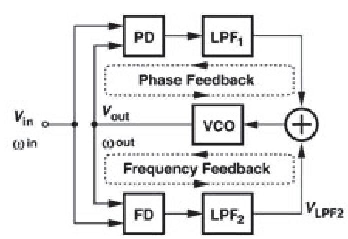
$$\text{Sideband Spur} \propto \omega_{LPF}$$

Limitations of Simple PLL (2)

- Limited acquisition range ~ ω_{LPF}
  - ✓ If the input frequency is too much far from the target frequency at startup, the loop is unlikely to lock forever, or possibly lock to a harmonic frequency of the input signal
  - ✓ Note: PD does not work for different frequency signals → When two input signals have different frequencies, the PD output will be a beat signal whose dc component is zero.

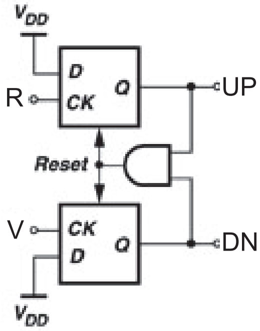


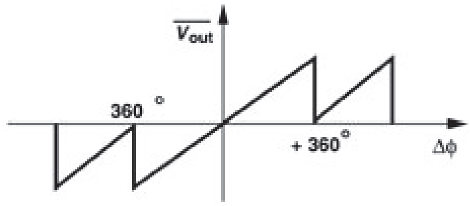
- Thus, additional frequency acquisition loop by using a FLL may be needed



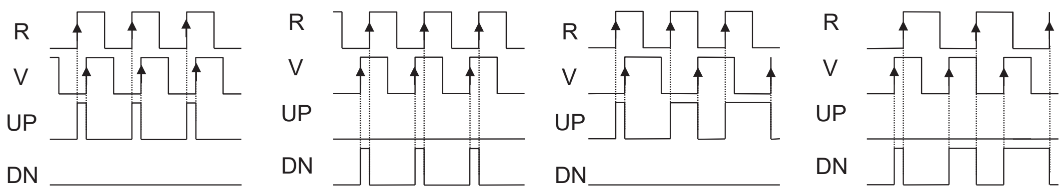
- *Charge-pump PLL can overcome these limitations.*
- *Charge-pump PLL* is the most popular PLL architecture in RF synthesizer due to *the far wider frequency and phase locking range*.

Tri-State Phase-Frequency Detector





Input-Output Transfer Characteristics



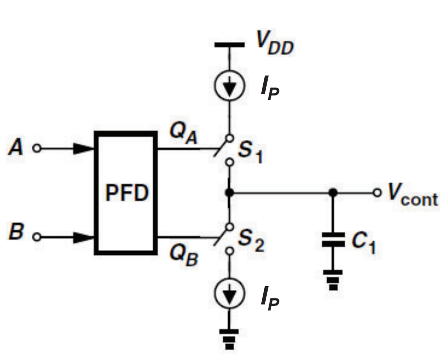
R leads V

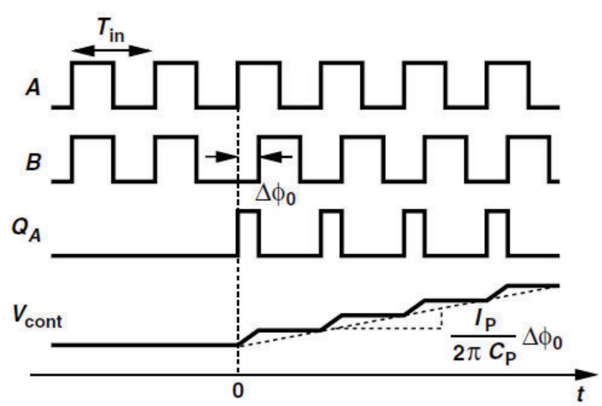
R lags V

$f_R > f_V$

$f_R < f_V$

PFD/CP/C_1 Transfer Function





The change in V_{cont} in every period

$$\Delta V_{cont} = \frac{\Delta\phi_o}{2\pi} \cdot T_{in} \cdot \frac{I_p}{C_1}$$

The slope of the ramp is given by

$$\frac{\Delta V_{cont}}{T_{in}} = \frac{\Delta\phi_o}{2\pi}\frac{I_p}{C_1}$$

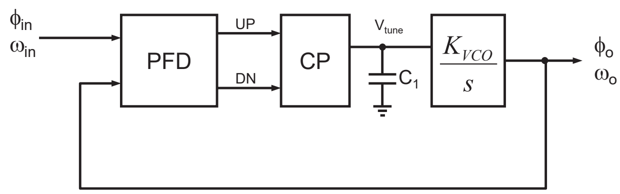
$V_{cont}(t)$ is given by

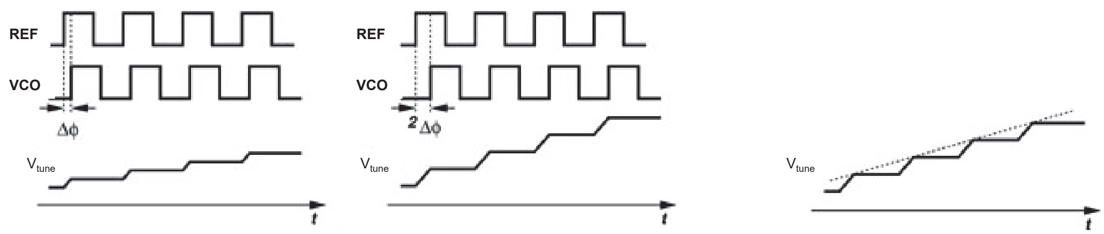
$$V_{cont}(t) \approx \frac{\Delta\phi_o}{2\pi}\frac{I_p}{C_1} t \cdot u(t)$$

Gain of PFD/CP/C_1 can be written as

$$\frac{V_{cont}}{\Delta\phi_o}(s) = \frac{I_p}{2\pi C_1}\frac{1}{s}$$

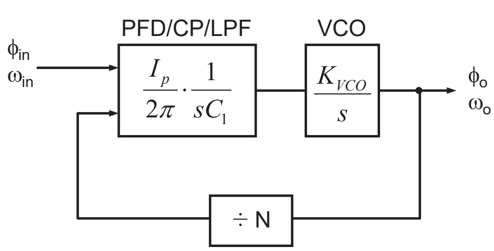
Waveforms in Charge-Pump PLL





Closed-Loop Transfer Function of CP PLL

- PLL is inherently a discrete-time system, working with a clock frequency fref
- But, we can model the discrete-time PLL as a continuous-time system
  - as long as Loop Bandwidth << fref → *What would happen if f_{REF} is not much larger than LBW ?*
- The following is the continuous-time model of the CP PLL



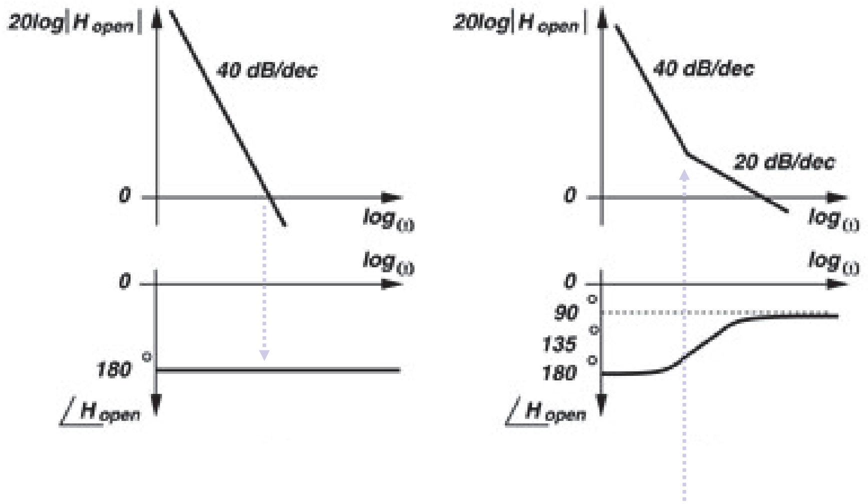
$$H(s)\Big|_{open} = \frac{\phi_{out}}{\phi_{in}}(s)\Big|_{open} = \frac{I_p}{2\pi C_1}\frac{K_{VCO}}{s^2}$$

→ named as Type-II because it has two poles at the origin.

$$H(s)\Big|_{closed} = N \cdot \frac{\frac{I_p K'_{VCO}}{2\pi C_1}}{s^2 + \frac{I_p K'_{VCO}}{2\pi C_1}}, where\ \ s_{p1,2} = \pm j\sqrt{\frac{I_p K'_{VCO}}{2\pi C_1}}, K'_{VCO} = \frac{K_{VCO}}{N}$$

→ Always unstable !!

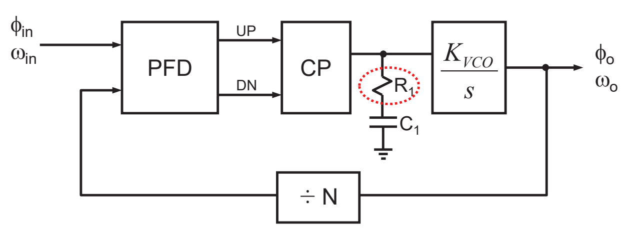
Stabilizing the Unstable CPPLL by adding a Zero



No phase-margin → unstable

Need a zero here !

Creating a Stabilizing Zero by adding a Resistor R_1



Transfer Function of 2nd-order CPPLL

$$H_{open}(s) = \frac{I_p K_{vco}}{2\pi C_1 s^2}(1 + R_1 C_1 s)$$

$$H_{closed}(s) = N \cdot \frac{\frac{I_p K'_{vco}}{2\pi C_1}(R_1 C_1 s + 1)}{s^2 + \frac{I_p K'_{vco}}{2\pi} R_1 s + \frac{I_p K'_{vco}}{2\pi C_1}}$$

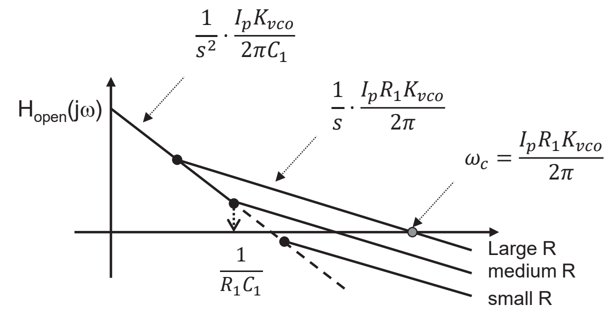
$$\omega_n = \sqrt{\frac{I_p K'_{VCO}}{2\pi C_1}}$$

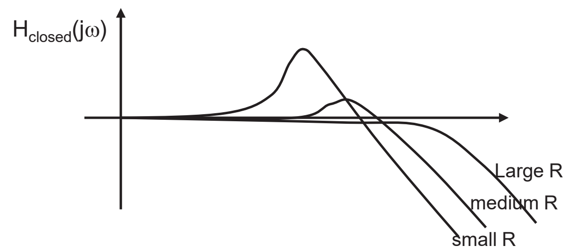
$$\zeta = \frac{R_1}{2}\sqrt{\frac{I_p C_1 K'_{VCO}}{2\pi}}$$

Stability Issue: R_1 Tuning

$$H_{open}(s) = \frac{I_p K_{vco}}{2\pi C_1 s^2}(1 + R_1 C_1 s)$$

$$H_{closed}(s) = N \cdot \frac{\frac{I_p K'_{vco}}{2\pi C_1}(R_1 C_1 s + 1)}{s^2 + \frac{I_p K'_{vco}}{2\pi} R_1 s + \frac{I_p K'_{vco}}{2\pi C_1}}$$

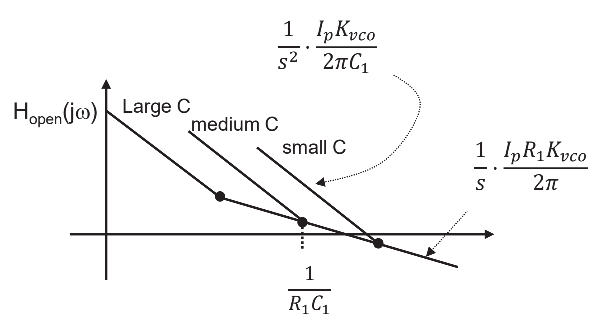


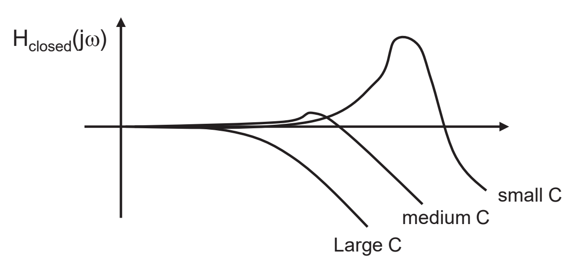


Stability Issue: C_1 Tuning

$$H_{open}(s) = \frac{I_p K_{vco}}{2\pi C_1 s^2}(1 + R_1 C_1 s)$$

$$H_{closed}(s) = N \cdot \frac{\frac{I_p K'_{vco}}{2\pi C_1}(R_1 C_1 s + 1)}{s^2 + \frac{I_p K'_{vco}}{2\pi} R_1 s + \frac{I_p K'_{vco}}{2\pi C_1}}$$

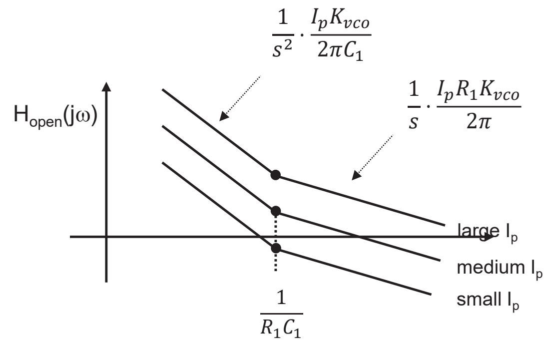


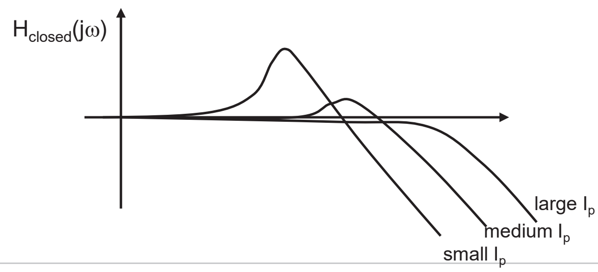


Stability Issue: I_p Tuning

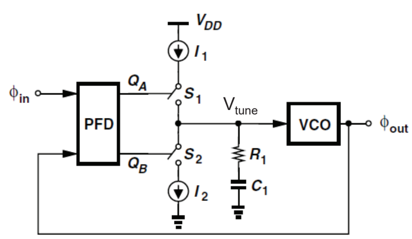
$$H_{open}(s) = \frac{I_p K_{vco}}{2\pi C_1 s^2}(1 + R_1 C_1 s)$$

$$H_{closed}(s) = N \cdot \frac{\frac{I_p K'_{vco}}{2\pi C_1}(R_1 C_1 s + 1)}{s^2 + \frac{I_p K'_{vco}}{2\pi} R_1 s + \frac{I_p K'_{vco}}{2\pi C_1}}$$





Transient Response

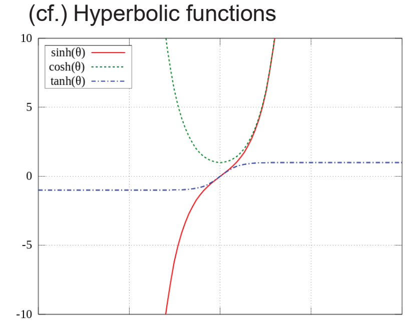


$$H(s) = \frac{\frac{I_p K_{vco}}{2\pi C_1}(R_1 C_1 s + 1)}{s^2 + \frac{I_p K_{vco}}{2\pi} R_1 s + \frac{I_p K_{vco}}{2\pi C_1}}$$

The inverse Laplace transform of the closed-loop transfer function gives

$$\Delta\omega_{out}(t) = \begin{cases} \Delta\omega_{in}u(t) - \Delta\omega_{in}\left[\cos\left(\sqrt{1-\zeta^2}\omega_n t\right) - \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right]e^{-\zeta\omega_n t}u(t)\ (\zeta < 1) \\ \Delta\omega_{in}u(t) - \Delta\omega_{in}(1-\omega_n t)e^{-\zeta\omega_n t}u(t)\quad (\zeta = 1) \\ \Delta\omega_{in}u(t) - \Delta\omega_{in}\left[\cosh\left(\sqrt{\zeta^2-1}\omega_n t\right) - \frac{\zeta}{\sqrt{\zeta^2-1}}\sinh\left(\sqrt{\zeta^2-1}\omega_n t\right)\right]e^{-\zeta\omega_n t}u(t)\ (\zeta > 1) \end{cases}$$

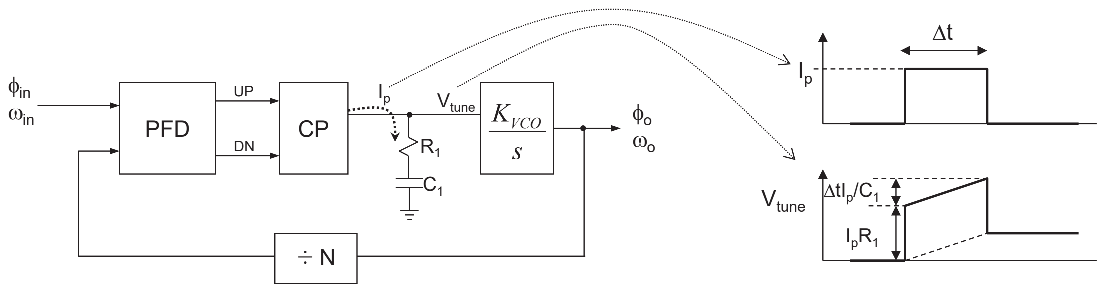
(cf.) Hyperbolic functions



- The loop time constant: $\frac{1}{\zeta\omega_n} = \frac{4\pi}{R_1 C_1 K_{VCO}}$

Ripple due to I_pR_1

- R_1 successfully stabilizes the loop.
- But R_1 creates unwanted ripple voltage on V_{tune} at the same time !!

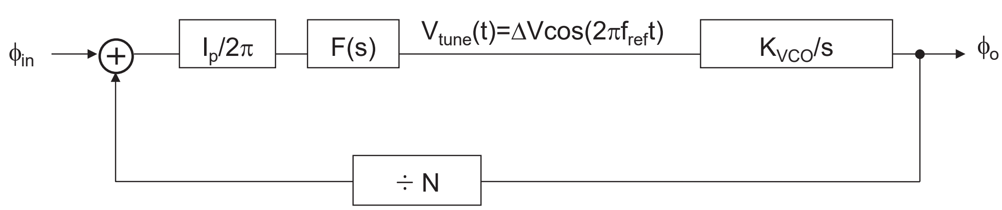


Periodically disturbing V_{tune} leads to creating large spur (a.k.a reference spur)

➔ Why? ➔ See the next slide

Ripple creates Sideband Spur

- Periodic fluctuation on the VCO tuning voltage generates sideband spurious tones

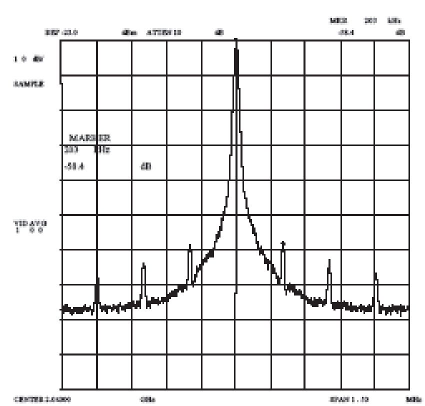


$$V_o(t) = A\cos\left[\omega_c t + 2\pi\cdot K_{VCO}\int V_{tune}(t)dt\right]$$

$$= A\cos\left[\omega_c t + \frac{K_{VCO}\cdot\Delta V}{f_{ref}}\sin\left(2\pi f_{ref}t\right)\right]$$

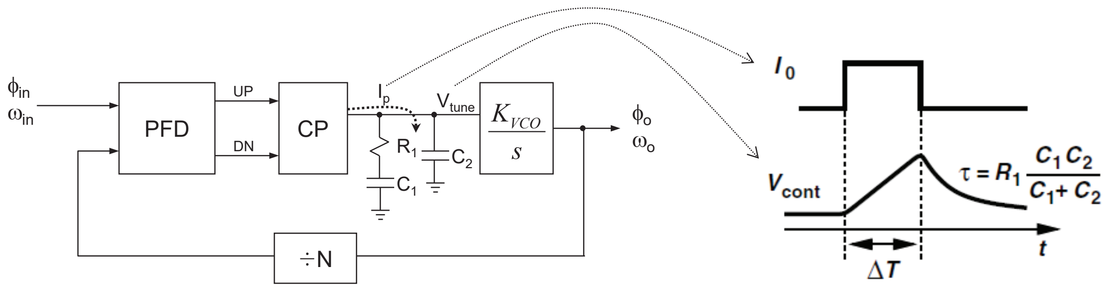
$$\cong A\cos(\omega_c t) \pm \frac{A}{2}\cdot\frac{K_{VCO}\cdot\Delta V}{f_{ref}}\cos\{2\pi\left(f_c \pm f_{ref}\right)t\}$$

$$\boxed{Spur = 20\log\left(\frac{1}{2}\cdot\frac{K_{VCO}\cdot\Delta V}{f_{ref}}\right)\quad \textbf{(dBc)}}$$



Higher-Order Loop to Reduce the Ripple

- Large voltage spike through R_1 & C_1 can be reduced by adding a small C_2

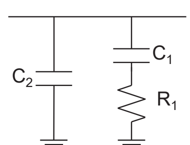


- Since C_2 poses an additional pole, it will potentially degrade the stability
- Practically, C_2 should be small enough in the range of

$$C_2 \approx (0.1 \sim 0.2) \cdot C_1$$

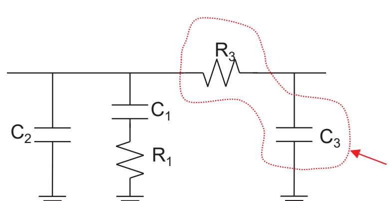
Higher-order Loop Filters

2nd-order Loop Filter → 3rd-order PLL



$$F(s) = \frac{1 + sC_1R_1}{s(C_1 + C_2)\left(1 + s\dfrac{C_1C_2R_1}{C_1 + C_2}\right)}$$

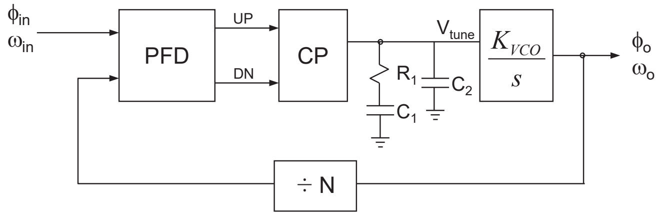
3rd-order Loop Filter → 4th-order PLL



Additional pole out of the loop bandwidth → additional attenuation

$$F(s) = \frac{1 + sC_1R_1}{s\left(C_1C_2C_3R_1R_3s^2 + s[C_1R_1(C_2 + C_3) + C_3R_3(C_1 + C_2)] + (C_1 + C_2 + C_3)\right)}$$

Type-II 3rd-order PLL: Transfer Function



$$Open\ Loop\ Gain\ G(s) = \frac{1}{N}\frac{I_p}{2\pi}\frac{K_{vco}}{s}\frac{1}{sC_2 + \frac{1}{R_1 + \frac{1}{sC_1}}} = \frac{I_p K_{vco}}{2\pi N}\frac{1 + sR_1C_1}{s^2(C_1 + C_2)\left(1 + sR_2\frac{C_1C_2}{C_1 + C_2}\right)}$$

$$Let\ \omega_z = \frac{1}{R_1C_1}, \omega_p = \frac{1}{R_1\frac{C_1C_2}{C_1 + C_2}},$$

$$\Rightarrow \boxed{Open - loop\ Gain\ G(s) = \frac{I_pK_{vco}}{2\pi N}\frac{1 + \frac{s}{\omega_z}}{s^2(C_1 + C_2)\left(1 + \frac{s}{\omega_p}\right)}}$$

$$\Rightarrow \boxed{Closed - Loop\ Gain\ H(s) = N\frac{G(s)}{1 + G(s)} = \frac{K_{cp}K_{vco}(1 + sR_1C_1)}{s^3R_1C_1C_2 + s^2(C_1 + C_2) + sR_1C_1\frac{K_{cp}K_{vco}}{N} + \frac{K_{cp}K_{vco}}{N}}}$$

Type-II 3rd-order PLL: Cross-over frequency ω_c

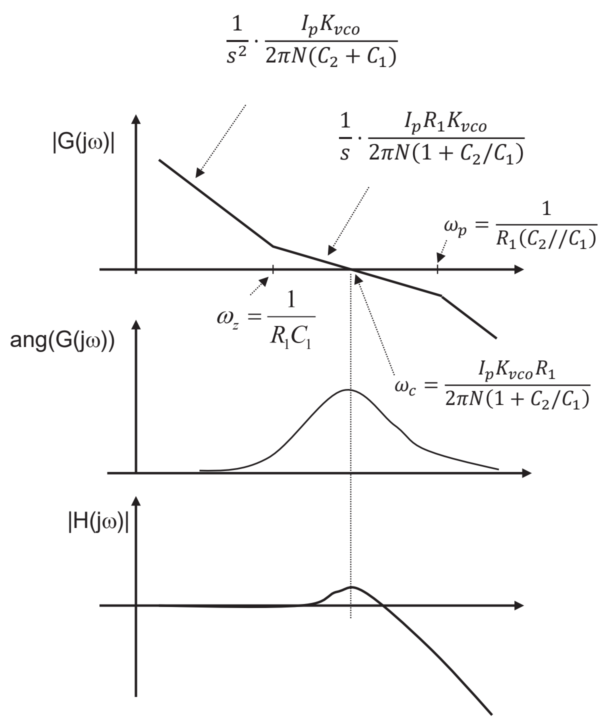
$$G(s) = \frac{I_pK_{vco}}{2\pi N}\frac{1 + \frac{s}{\omega_z}}{s^2(C_1 + C_2)\left(1 + \frac{s}{\omega_p}\right)}$$

$$if\ \omega_z << \omega_c << \omega_p$$

$$|G(j\omega_c)| = \frac{I_pK_{vco}}{2\pi N}\left|\frac{1 + \frac{j\omega_c}{\omega_z}}{\omega_c^2(C_1 + C_2)\left(1 + \frac{j\omega_c}{\omega_p}\right)}\right|$$

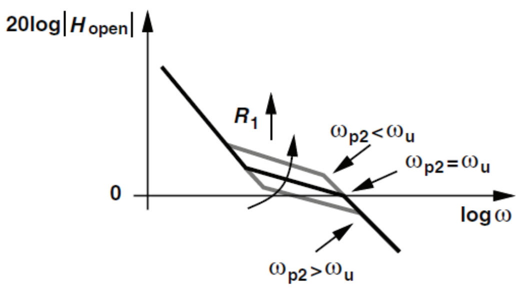
$$\approx \frac{I_pK_{vco}}{2\pi N}\frac{\frac{\omega_c}{\omega_z}}{\omega_c^2(C_1 + C_2)} = 1$$

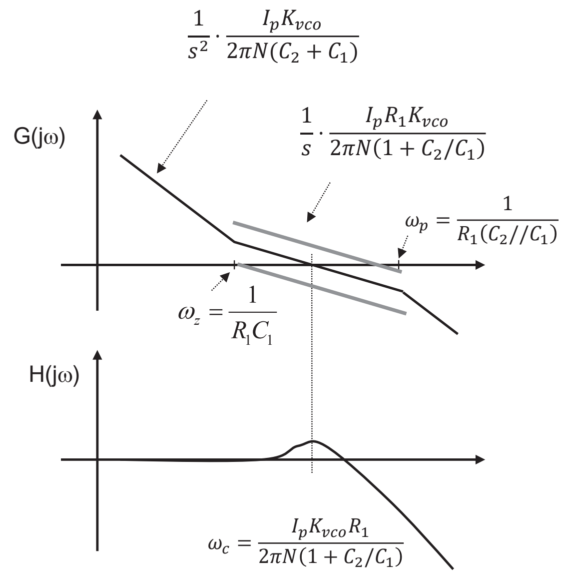
$$\therefore\ \omega_c = \frac{I_pK_{vco}}{2\pi N}\frac{R_1C_1}{(C_1 + C_2)} \approx \frac{I_pK_{vco}R_1}{2\pi N}$$



Effect of R_1 on the Loop Stability

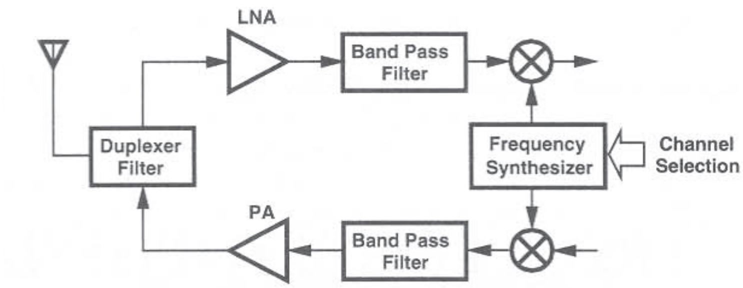
Too much high or low R_1 will make the loop unstable.
After all, both cases will have the same effect as removing R_1.





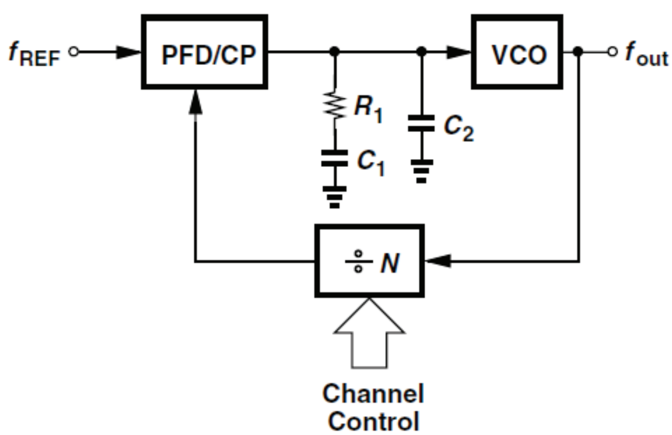
Integer-N Frequency Synthesizers

Frequency Synthesizer in RF Transceiver



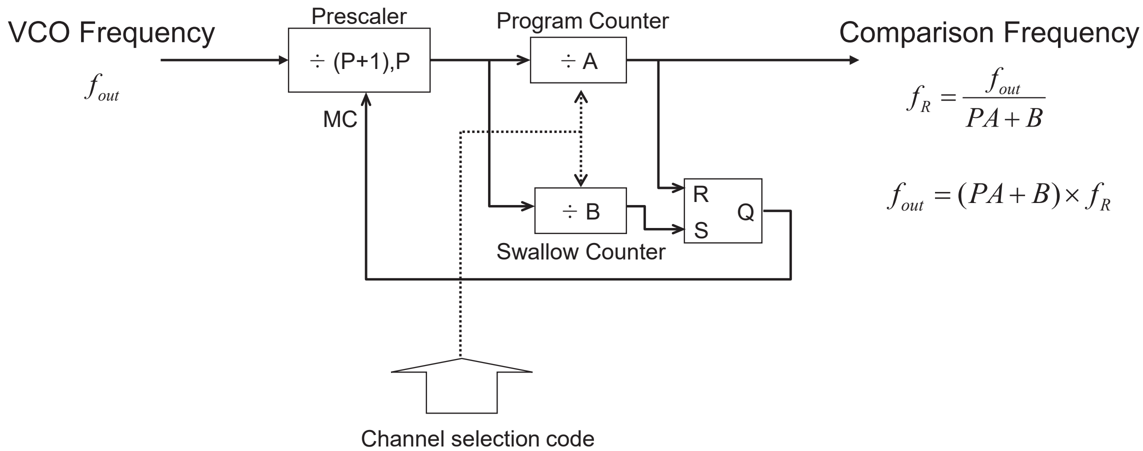
- Receiver:
  - LNA and Down-conversion mixer
- Transmitter:
  - Up-conversion mixer and power amplifier
- PLL:
  - Synthesizer provides LO signal for the up/down-conversion mixers

Basic Integer-N Synthesizer



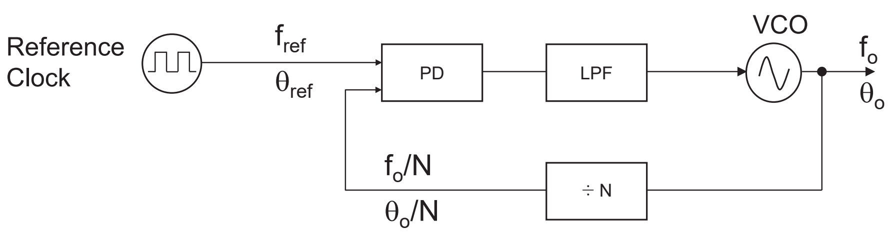
$$f_{out} = N \cdot f_{REF}$$

Pulse Swallow Divider



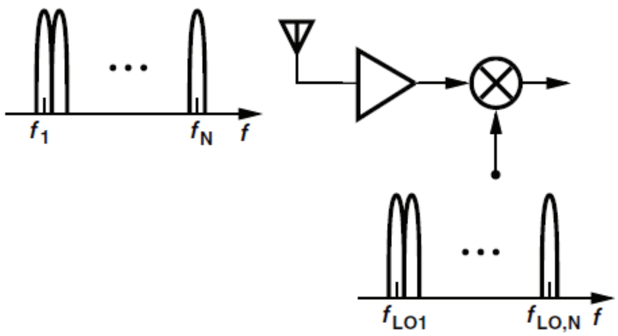
Total Divide Ratio (N) = (P+1)×B + P×(A-B) = P·A+B

Frequency Coverage and Channel Spacing



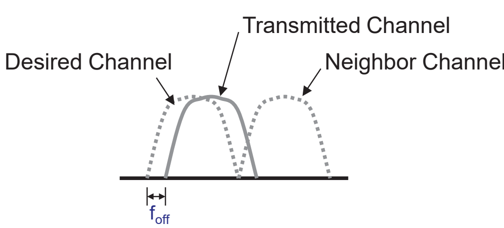
- Frequency coverage is determined by the VCO tuning range and the divide-ratio range.
  - $\{f_{o,max} \sim f_{o,min}\}$ and $\{(N_{max} \sim N_{min}) \times f_{ref}\}$
- Channel Spacing
  - Integer-N Frequency Synthesizer
    - f_{ref} because $f_o = N \times f_{ref}$ where N is an integer number
  - Fractional-N Frequency Synthesizer
    - smaller than f_{ref} because $f_o = N.f \times f_{ref}$ where (N.f) is a fractional number

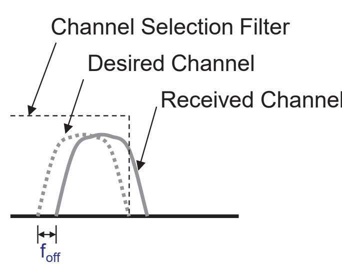
Channel Selection in Receiver



- LO frequency must be properly set to select a wanted channel among the multiple channels.
- Frequency synthesizer is needed for this purpose.

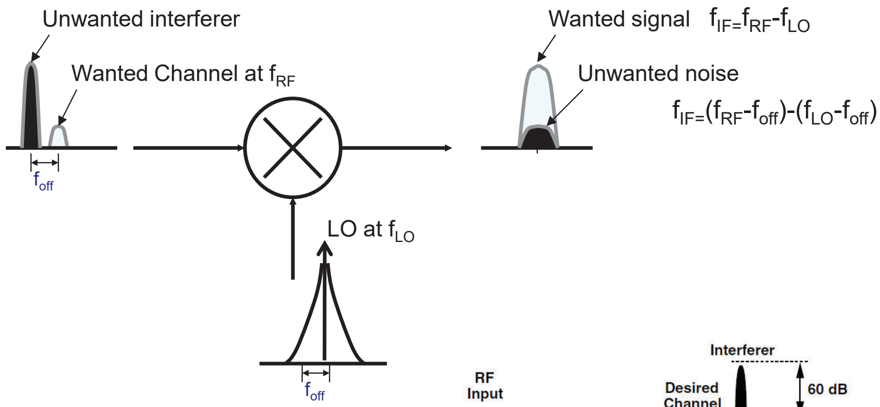
Effect of LO Frequency Error

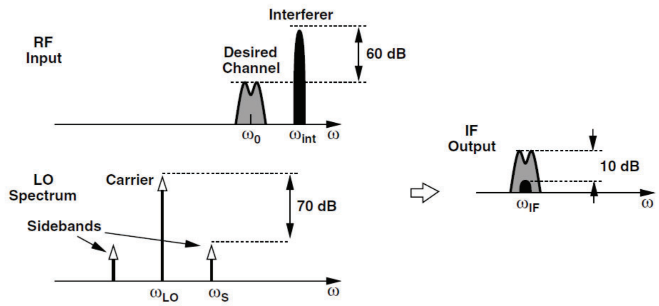




- Frequency accuracy is dictated by the communication standards (e.g. ±150Hz for IS-95B)
- It can destroy a neighbor channel in a transmitter, and cause a partial loss of a received signal in a receiver.
- Frequency accuracy is mainly determined by the accuracy of the reference clock frequency.
  - e.g. if TCXO with 60 ppm accuracy at 20 MHz is used for 2.4 GHz FS, the frequency accuracy will be 60 ppm x 2.4GHz = 144 kHz

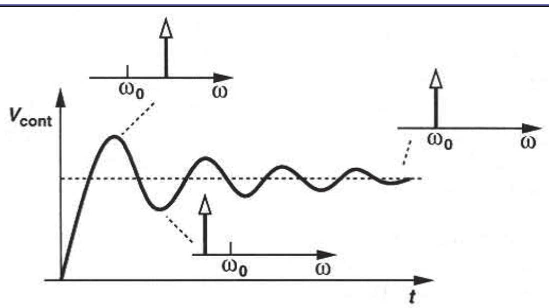
Reciprocal Mixing

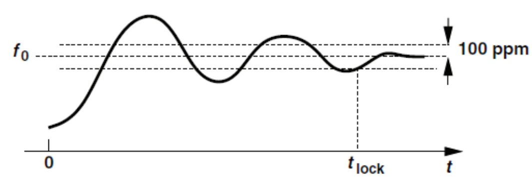




• SNR degradation (dB) = Interferer (dBc) – Phase Noise (dBc)

Loop Transient Behavior – Lock Time





- Settling time requirements are set by communication standards;
  - inter-frequency hand-off or hard hand-off
  - intermittent power on/off for power saving management
  - frequency hopping (IEEE 802.15.4 or IEEE 802.11g)
- Lock time
  - defined by the time needed to switch the frequency with a certain accuracy (*e.g.* 150 ms with an accuracy of ±100ppm)
  - mainly determined by the feedback loop dynamics (loop bandwidth and phase margin)
  - significantly affected by unexpected non-linear disturbances such as switching noise.

Settling Behavior

- In RF synthesizer, the transients of interest are
  (1) Startup switching: the synthesizer has been off and now switched on
  (2) Channel switching: the feedback divide ratio changes in order to hop from one channel to another → equivalent to the case when the input frequency changes

$$Y(s) = \frac{H(s)}{1 + (A + \varepsilon)H(s)} X(s)$$

$$\approx \frac{H(s)}{1 + AH(s)} \cdot \frac{1}{1 + \varepsilon/A} X(s)$$

$$\approx \frac{H(s)}{1 + AH(s)} \left(1 - \frac{\varepsilon}{A}\right) X(s)$$

Settling Time Calculation

$$\left|1 - \frac{N_1}{N_2}\right| g(t)u(t) \le \alpha$$

where

$$g(t) = 1 - \left[\cos\left(\sqrt{1-\zeta^2}\omega_n t\right) - \frac{\zeta}{\sqrt{1-\zeta^2}} \sin\left(\sqrt{1-\zeta^2}\omega_n t\right)\right] e^{-\zeta\omega_n t} \quad (\zeta < 1)$$

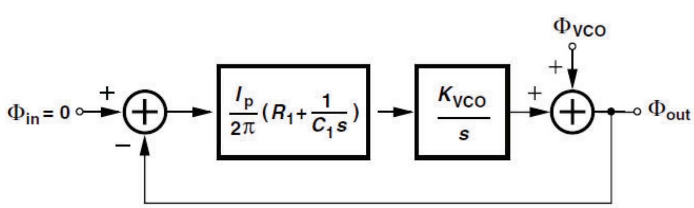
$$= 1 - (1 - \omega_n t)e^{-\zeta\omega_n t} \quad (\zeta = 1)$$

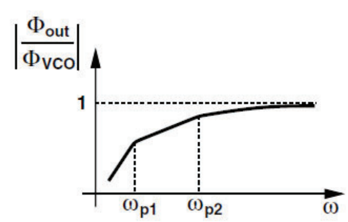
$$= 1 - \left[\cosh\left(\sqrt{\zeta^2 - 1}\omega_n t\right) - \frac{\zeta}{\sqrt{\zeta^2 - 1}} \sinh\left(\sqrt{\zeta^2 - 1}\omega_n t\right)\right] e^{-\zeta\omega_n t} \quad (\zeta > 1)$$

For example, when $\zeta = \frac{\sqrt{2}}{2} \Rightarrow t_s = \frac{\sqrt{2}}{\omega_n} \ln\left|\sqrt{2}\left(1 - \frac{N_1}{N_2}\right)\frac{1}{\alpha}\right|$

PLL Design Issues:
Noise and Non-idealities

VCO Phase Noise at PLL Output

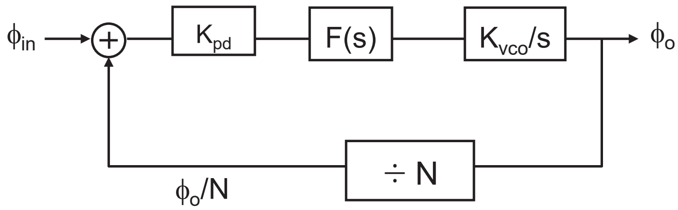




$$\frac{\phi_{out}}{\phi_{VCO}} = \frac{s^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

→ exhibiting high-pass behavior

Phase Noise in PLL

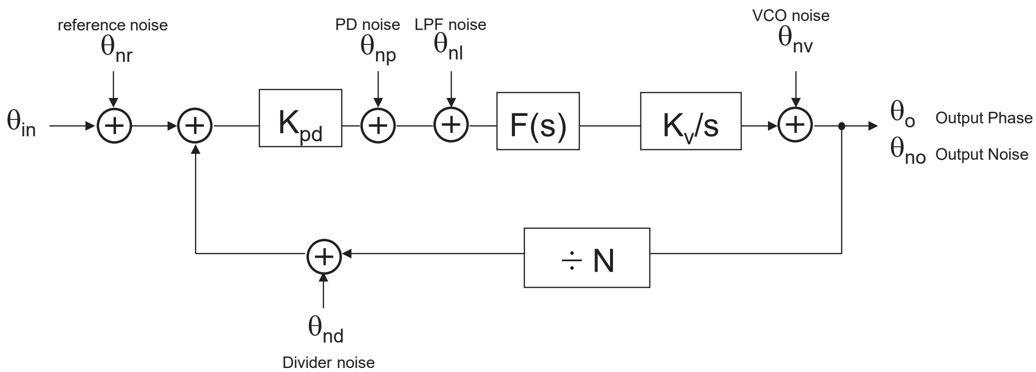


- $Open\ Loop\ Transfer\ Function\ G(s) = \dfrac{K_{pd}K_{vco}F(s)}{s}$
- $Closed\ Loop\ Transfer\ Function\ H(s) = \dfrac{\phi_o(s)}{\phi_{in}(s)} = \dfrac{G(s)}{1 + \frac{1}{N}\cdot G(s)}$

(cf.) If F(s) is a first−order filter,

H(s) becomes second−order system as such $H(s) = \dfrac{{\omega_n}^2}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$

Noise Analysis of PLL

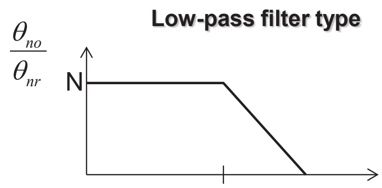


$$\theta_{no}^2 = \left(\theta_{nr}^2 + \theta_{nd}^2 + \frac{(\theta_{np}^2 + \theta_{nl}^2)}{K_{pd}^2}\right)\cdot\left(\frac{G(s)}{1 + G(s)/N}\right)^2 + \theta_{nv}^2\cdot\left(\frac{1}{1 + G(s)/N}\right)^2$$

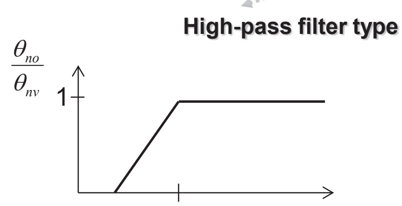
For the 2nd-order PLL; $\dfrac{G(s)}{1 + G(s)/N} = \dfrac{{\omega_n}^2}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$ $\dfrac{1}{1 + G(s)/N} = \dfrac{s^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + {\omega_n}^2}$

Noise Transfer Characteristics

$$\theta_{no}^2 = \left(\theta_{nr}^2 + \theta_{nd}^2 + \frac{(\theta_{np}^2 + \theta_{nl}^2)}{K_{pd}^2}\right) \cdot \left(\frac{G(s)}{1+G(s)/N}\right)^2 + \theta_{nv}^2 \cdot \left(\frac{1}{1+G(s)/N}\right)^2$$



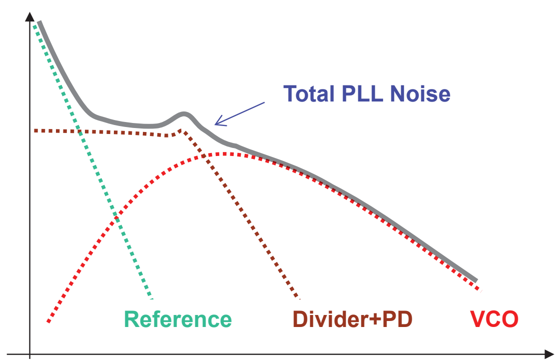
Output tracks slow input phase variations



Output tracks fast VCO phase variations

- In-band Phase Noise
  - Approximately, Phase Detector Noise Floor (PDNF) + 20log(N)
  - *ex)* PDNF = – 150 dBc/Hz, f_{ref} = 1 MHz, f_{out} = 2 GHz, N = 2000 → −150+20log(2000) = – 84dBc/Hz
- Out-of-band Phase Noise
  - Almost equal to VCO Phase Noise

Typical Phase Noise Characteristics of PLL



- In-band noise is proportional to the divide ratio (N^2)

$$\theta_{no}^2 \propto \theta_{nr}^2 \cdot \left(\frac{G(s)}{1+G(s)/N}\right)^2 \approx N^2 \cdot \theta_{nr}^2$$

- Out-of-band noise follows the inherent VCO phase noise
- Possible noise peaking due to the loop transfer characteristics

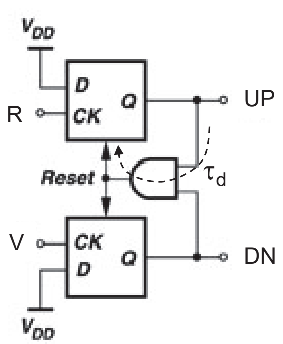
PFD and CP Design

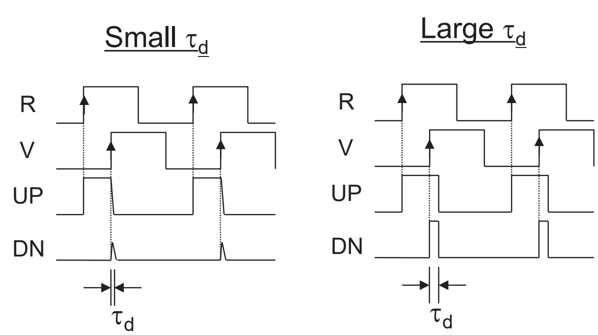
- PFD/CP
  - converts the phase error between ϕ_{ref} and ϕ_{vco} into a linearly proportional current and subsequently into a voltage

$$K_{PFD+CP} = \frac{V_{tune}}{\Delta\varphi}(s) = \frac{I_p}{2\pi} \cdot F(s)$$

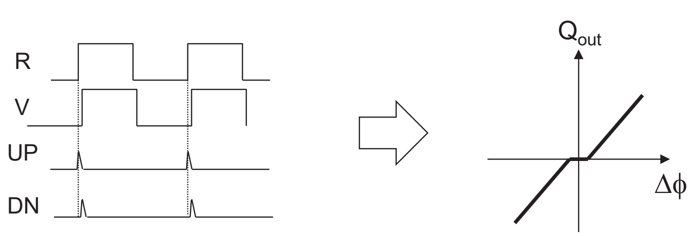
- Design Issues
  - Gain (~I_p)
  - Linearity
  - Dead-zone
  - Mismatch: Up/Down Current Mismatch, Up/Down Pulse Timing Mismatch
  - Off-state leakage current
  - Charge injection

PFD Dead-Zone Issue (1)

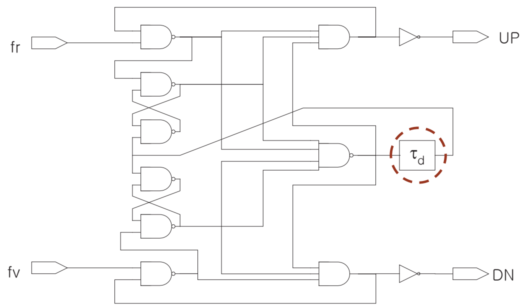




When the phase difference is small, and if the reset delay (τ_d) is also small,
$Q_{UP\ \&\ DN}$ will be reset too fast even before it can completely build up a pulse whose width is proportional to $\Delta\phi$.

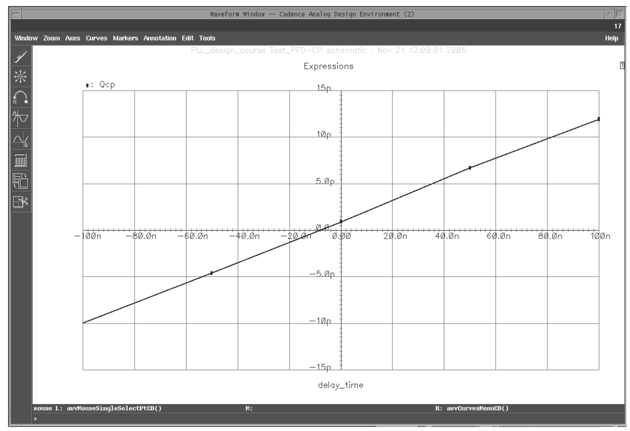


PFD Dead-Zone Issue (2)

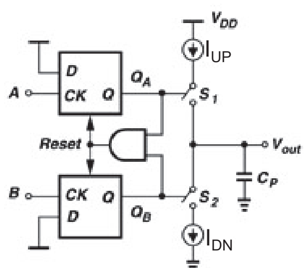


❖ Dead-zone elimination technique
- Increasing τ_d to guarantee a minimum pulse width
- which will lead to the limitation of the maximum operating speed of PFD

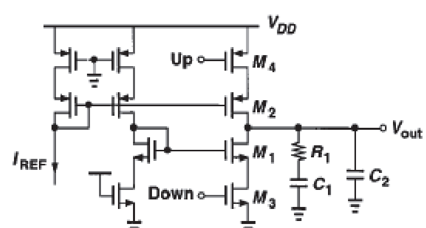
❖ PFD+CP Transfer Characteristic Simulation Result



Charge-Pump Circuit



Charge-Pump Circuit Example

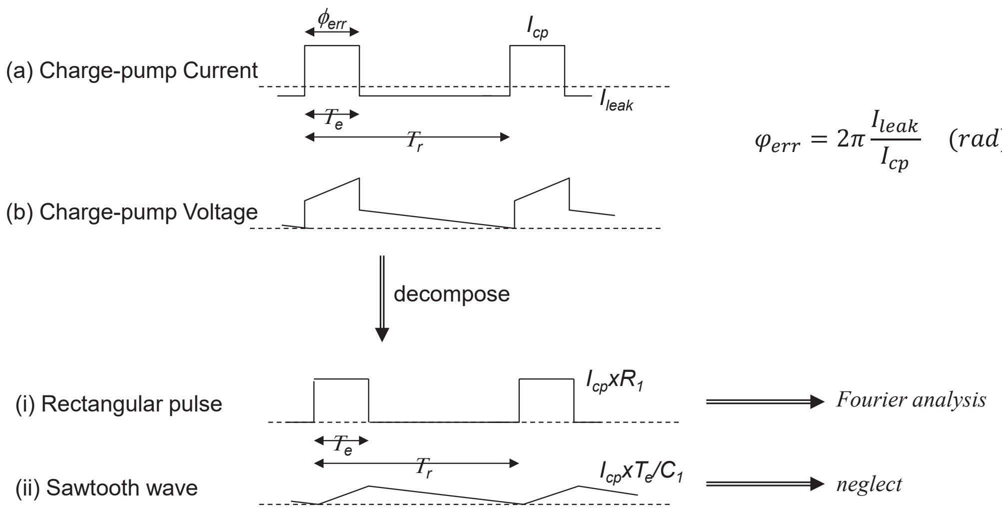


- M1/M2 generate UP/DN current
- M3/M4 act as switches

❖ Charge Pump Circuit Design Issues
- Off-state Leakage current
- Up/Down Current Mismatch
- Timing Mismatch (Pulse Skew)
- Charge Injection
- Voltage compliance

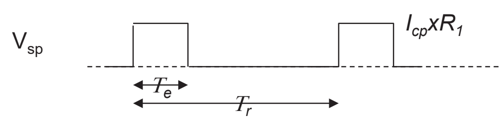
Leakage Current in Charge Pump Circuit

- Leakage Current: Off-state leakage current of FET switch causes phase offset



$$\varphi_{err} = 2\pi \frac{I_{leak}}{I_{cp}} \quad (rad)$$

Fourier Analysis of Ripple Voltage due to Leakage



Take the Fourier series expansion of the above rectangular waveform,

$$V_{sp} = \sum_{n=1}^{\infty} \frac{2}{n\pi} I_{cp} R_1 \sin\left(\frac{T_e}{T_r} n\pi\right) \cos(2\pi n f_r t)$$

Take its fundamental component

$$V_{sp,1} = \frac{2}{\pi} I_{cp} R_1 \sin\left(\frac{T_e}{T_r}\pi\right) \cos(2\pi f_r t) \cong 2 I_{cp} R_1 \frac{T_e}{T_r} \cos(2\pi f_r t)$$

$$V_{sp,1} = 2 I_{leak} R_1 \cos(2\pi f_r t)$$

Plug it into the VCO expression and calculate the sideband spur,

$$\begin{aligned} v_{RF} &= \cos\left(2\pi K_{vco} \int (V_o + V_{sp,1}) dt\right) \\ &= \cos\left(2\pi K_V \int V_o dt + 2\pi K_{vco} \int V_{sp,1} dt\right) \\ &= \cos\left(\omega_{RF} t + \frac{2 I_{leak} R_1 K_{vco}}{f_r} \sin(2\pi f_r t)\right) \end{aligned}$$

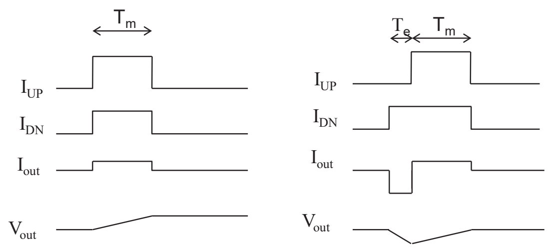
Calculate spur considering the additional loop attenuation,

$$\Rightarrow \quad P_{spur} = 20\log\left(\frac{I_{leak} \cdot R_1 \cdot K_{VCO}}{f_{ref}}\right) - 20\log\left(\frac{f_{ref}}{f_{p1}}\right) \quad \text{(dBc)}$$

Current Mismatch in Charge Pump Circuit

❖ Current Mismatch

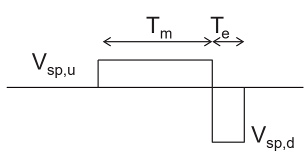
- Mismatches between UP/DOWN current creates unwanted ripple and spurs
- The amount of mismatch can vary with the output voltage



T_m: minimum pulse width

$$\frac{T_e}{T_m} = \frac{\Delta I_{cp}}{I_{cp}}$$

Fourier Analysis of Ripple Voltage due to Mismatch



Take the Fourier series expansion of the upper and lower rectangular waveform,

$$V_{sp,u} = \sum_{n=1}^{\infty} \frac{2}{n\pi} \Delta I_{cp} R_1 \sin\left(\frac{T_m}{T_r} n\pi\right) \cos(2\pi n f_r t)$$

$$V_{sp,d} = \sum_{n=1}^{\infty} \frac{2}{n\pi} I_{cp} R_1 \sin\left(\frac{T_e}{T_r} n\pi\right) \cos\left(2\pi n f_r \left(t - \frac{T_e + T_m}{2}\right)\right)$$

Take their fundamental components,

$$V_{sp,u1} = 2\Delta I_{cp} R_1 \frac{T_m}{T_r} \cos(2\pi f_r t)$$

$$V_{sp,d1} = 2\Delta I_{cp} R_1 \frac{T_m}{T_r} \left[\cos(2\pi f_r t) - 2\pi \cdot \frac{T_e + T_m}{2T_r} \sin(2\pi f_r t)\right]$$

Calculate $V_{sp,u1}$-$V_{sp,d1}$

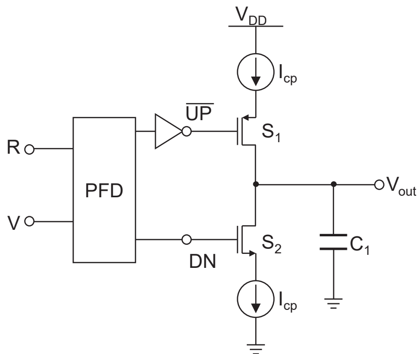
$$V_{sp1} = 2\pi \Delta I_{cp} R_1 \left(\frac{T_m}{T_r}\right)^2 \left(1 + \frac{\Delta I_{cp}}{I_{cp}}\right) \sin(2\pi f_r t)$$

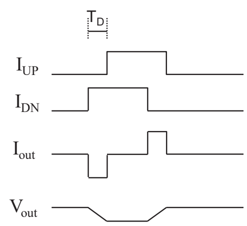
Calculate spur considering the additional loop attenuation,

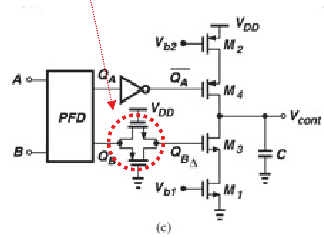
$$P_{spur} = 20 \log\left(\frac{\pi \cdot \Delta I_{cp} \cdot R_1 \cdot K_V \cdot \left(T_m f_{ref}\right)^2 \left(1 + \frac{\Delta I_{cp}}{I_{cp}}\right)}{f_{ref}}\right) - 20 \log\left(\frac{f_{ref}}{f_{p1}}\right) \quad \text{(dBc)}$$

Timing Mismatch in Charge Pump Circuit

❖ Timing Mismatch
- Delay difference between UP/DOWN pulse creates unwanted ripple and spur
- Both delays can be equalized by using a dummy pass-gate







Charge Injection & Clock Feedthrough

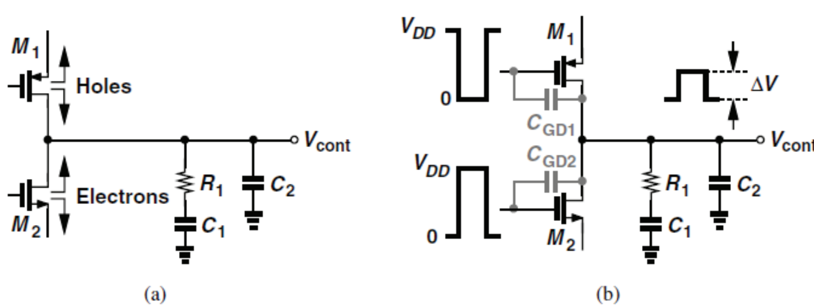


Figure 9.45 *(a) Channel charge injection, and (b) clock feedthrough in a charge pump.*

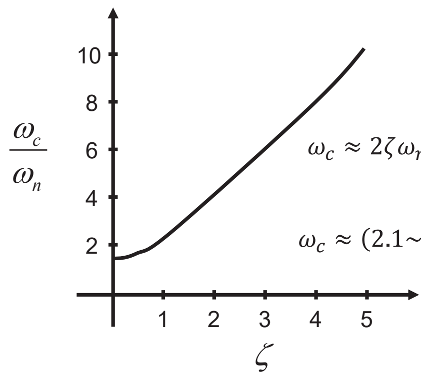
❑ Charge injection
- Channel charge in $M_{1,2}$ disturbs V_{cont} when they turn on and off.

❑ Clock feedthrough
- Large switching voltage disturbs V_{cont} through the gate-drain overlap capacitances of $M_{1,2}$

Loop Bandwidth (1)

From the second-order type-II PLL closed-loop transfer function

$$|H(j\omega_c)| = \left| N \cdot \frac{\frac{I_p K'_{vco}}{2\pi C_1}(jR_1 C_1 \omega_c + 1)}{(j\omega_c)^2 + j\frac{I_p K'_{vco}}{2\pi} R_1 \omega_c + \frac{I_p K'_{vco}}{2\pi C_1}} \right| = \frac{N}{2} \quad \Rightarrow \quad \omega_c = \omega_n \left[2\zeta^2 + 1 + \sqrt{(2\zeta^2+1)^2 + 1} \right]^{1/2}$$



$\omega_c \approx 2\zeta\omega_n$ when $\zeta >> 1$

$\omega_c \approx (2.1\sim2.5) \times \zeta\omega_n$ when $\zeta = 0.707\sim1$

Loop Bandwidth (2)

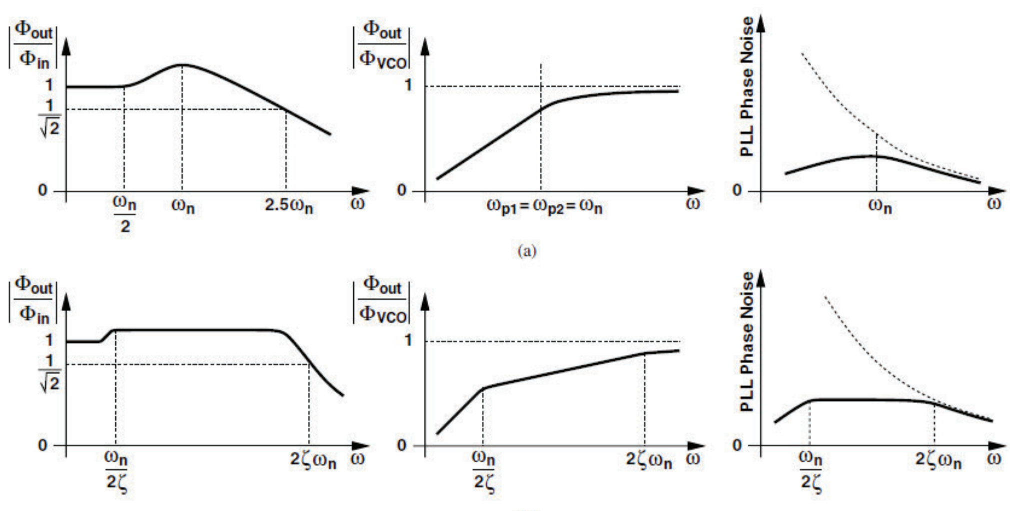
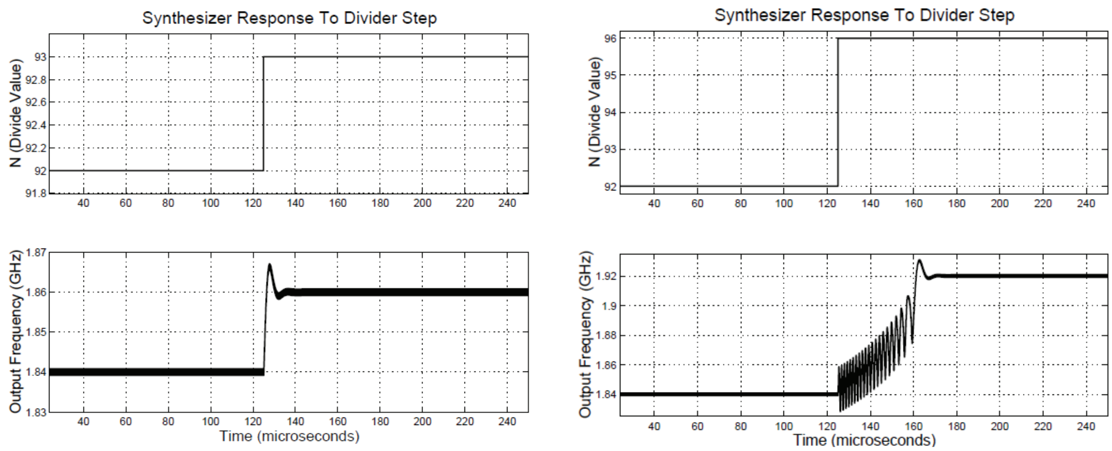
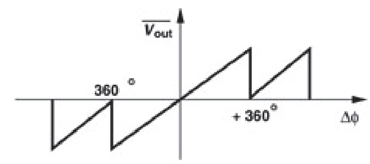


Figure 9.65 *Frequency responses and shaped VCO phase noise for (a) $\zeta = 1$, and (b) $\zeta^2 \gg 1$.*

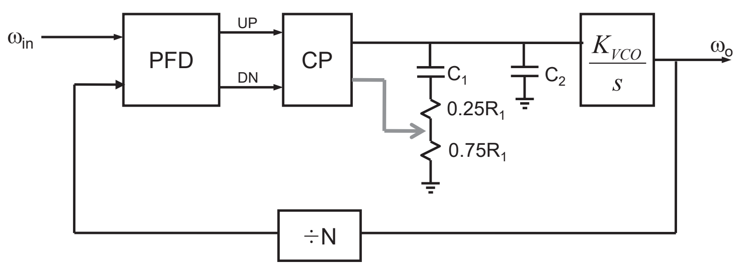
Cycle Slip



PLL temporarily loses frequency lock (cycle slip)
Why does this happen ?
Because the highly nonlinear PFD range



Fast Lock Technique



In fast lock mode, damping resistor is reduced to one fourth, and I_p increases by a factor of 16

→ then, loop bandwidth is widened x4 while the damping factor remains unchanged

cf. Transfer function of 2nd-order PLL

$$H(s) = N \cdot \frac{\frac{I_p K'_{vco}}{2\pi C_1}(R_1 C_1 s + 1)}{s^2 + \frac{I_p K'_{vco}}{2\pi} R_1 s + \frac{I_p K'_{vco}}{2\pi C_1}}$$

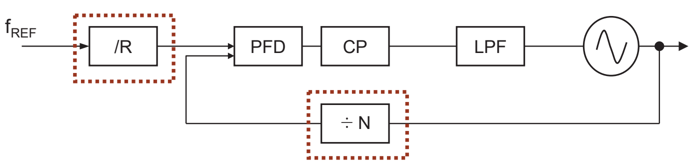
$$\omega_n = \sqrt{\frac{I_p K'_{VCO}}{2\pi C_1}}$$

$$\zeta = \frac{R_1}{2}\sqrt{\frac{I_p C_1 K'_{VCO}}{2\pi}}$$

Frequency Dividers

- Pulse Swallow Divider
- Dual-Modulus Divider
- Divider Logic Style:
  - CML & TSPC

Frequency Dividers in Synthesizer



$$f_{out} = N \times \frac{f_{REF}}{R}$$

❖ Reference-Path Divider

- Reference frequency is usually from TCXO with < 50MHz
- Programmable to accommodate various channel spacing or comparison freq.
- Synchronous type is preferred to a ripple counter type for the low phase noise

❖ Feedback-Path Divider

- Divider modulus N must change in unity step
- The first stage of the divider must handle the fast VCO signal (speed issue)
- Required input swing of the divider must be commensurate with the VCO driving capability (swing issue)
- Low power consumption (power issue)
- The most common realization is the pulse swallow divider

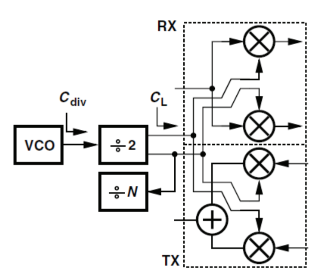
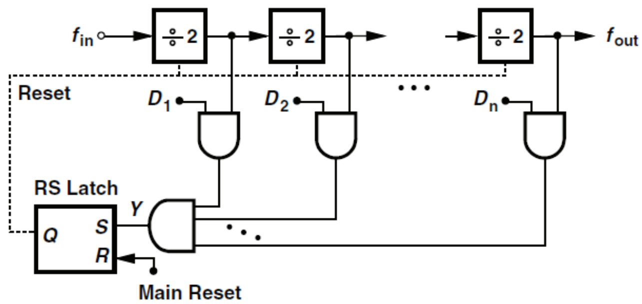
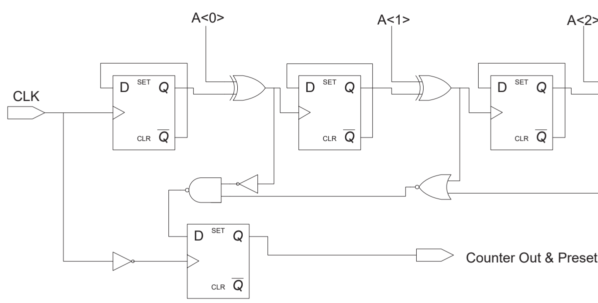


Figure 10.23 *Load seen by divider in a transceiver.*

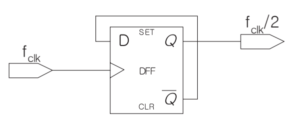
Programmable Ripple Counter

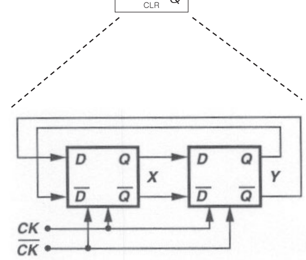




→ *Only NAND and NOR gates are used.*

Divide-by-2 Circuit

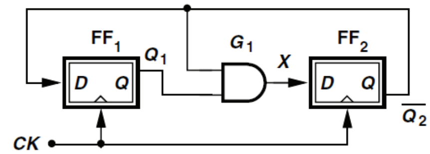




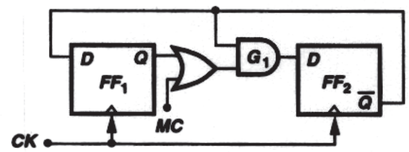
- Divide-by-2 is usually constructed by using Master-slave D flip-flop with the QB connected to the D-input

Dual-Modulus Divider (1)

❖ Divide-by-3 Circuit

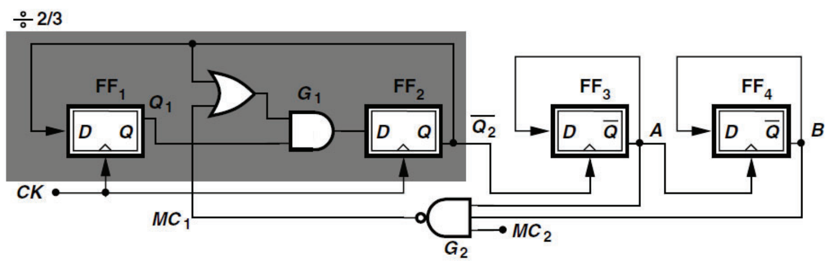


❖ Divide-by-2/3 Circuit



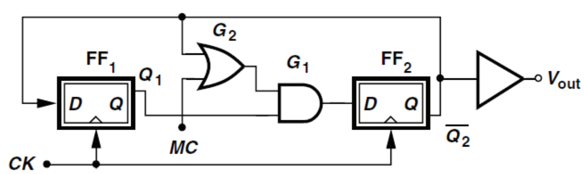
• MC=0 → Divide by 3
• MC=1 → Divide by 2

❖ Divide-by-8/9 Circuit

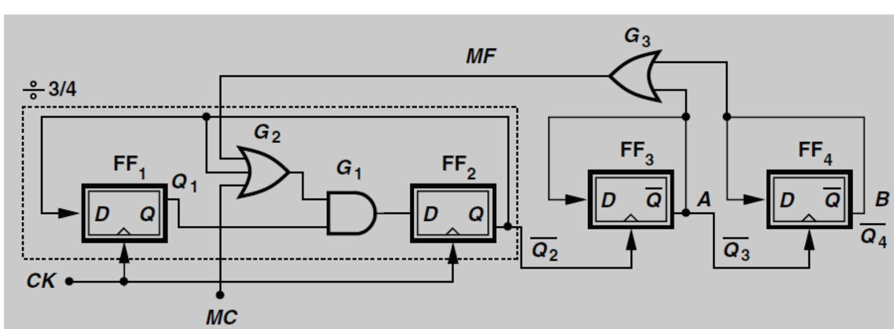


Dual-Modulus Dividers (2)

❖ Divide-by-3/4 Circuit

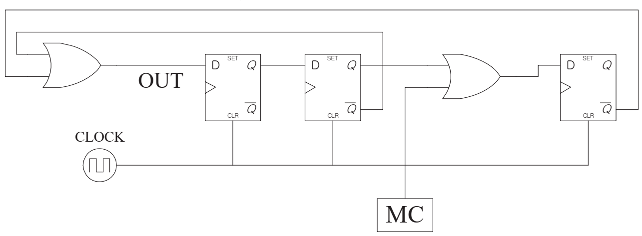


❖ Divide-by-15/16 Circuit

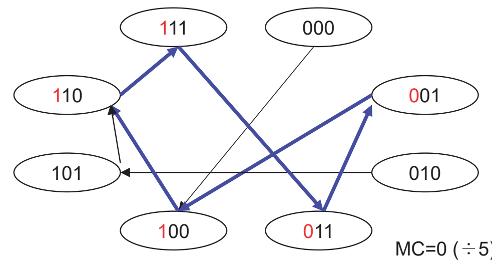


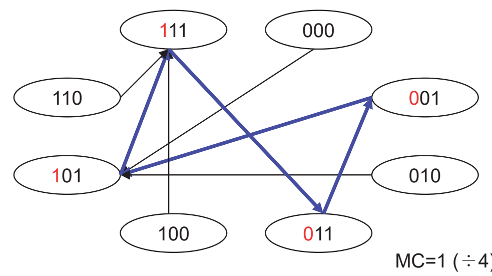
Dual-Modulus Dividers (3)

Divide-by-4/5



• MC=0 → Divide by 5
• MC=1 → Divide by 4

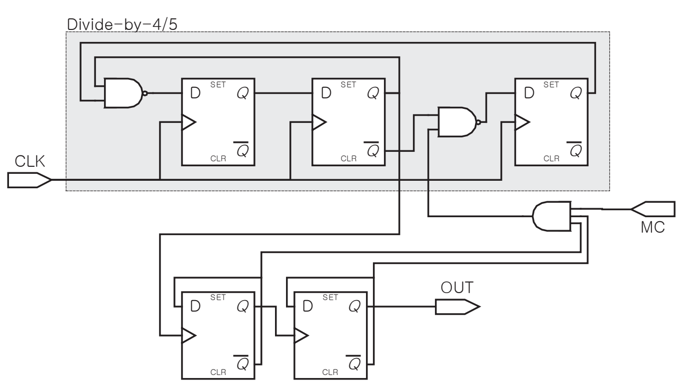




Current state			Next state			
			$D_A=\underline{Q_B+Q_C}$	$D_B=Q_A$	$D_C=MC+Q_B$	
Q_A	Q_B	Q_C			MC=0 (÷5)	MC=1 (÷4)
0	0	0	1	0	0	1
0	0	1	1	0	0	1
0	1	0	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	1	0	1
1	0	1	1	1	0	1
1	1	0	1	1	1	1
1	1	1	0	1	1	1

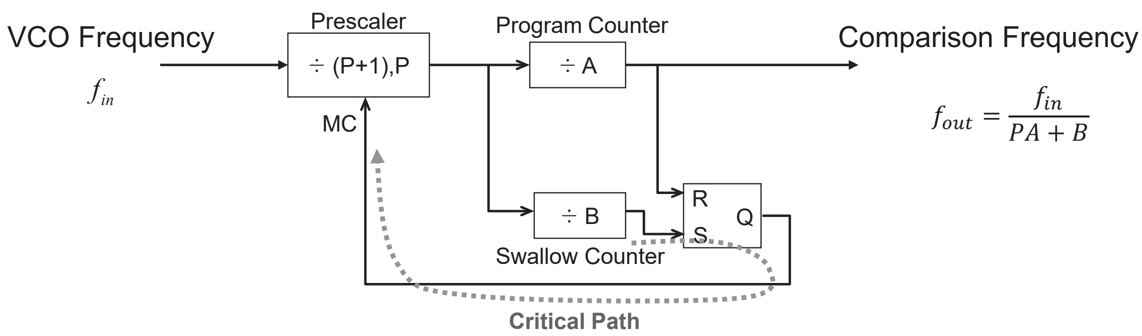
Dual-Modulus Divider (4)

❖ Dual-Modulus Prescaler (Divide-by-16/17)



- Dual-modulus prescaler extends a synchronous div-4/5 with following two div-2's
- Each 1/2-divider increase the divide-ratio by a factor of 2
- Maximum speed is determined in the critical MC path of the synchronous counter

Choice of Prescaler Modulus



- Prescaler modulus must be chosen carefully considering the speed and power trade-off
- Higher P lowers the critical path delay issue, and increases the power consumption of prescaler

Divider Logic Style: CML (1)

❖ Current Steering Logic, aka Current-Mode Logic (CML)

CML NAND Circuit

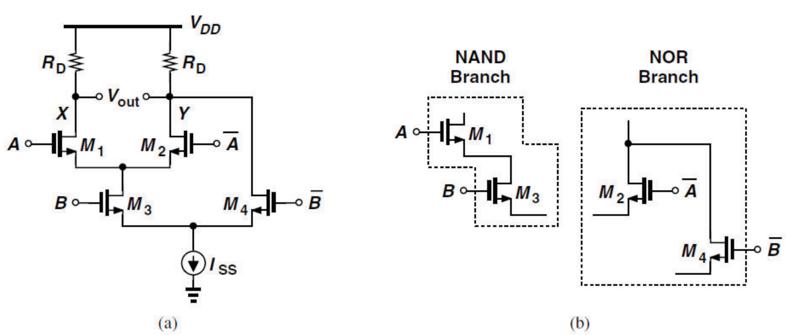


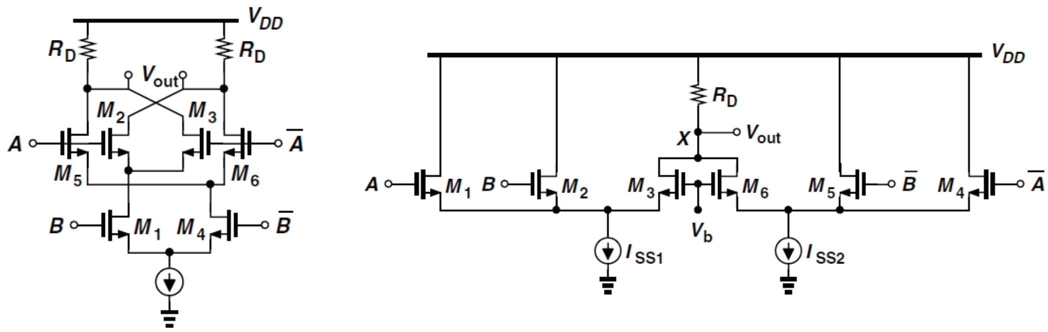
Figure 10.38 *(a) CML NAND realization, (b) NAND and NOR branches in the circuit.*

$$X = \overline{AB}$$

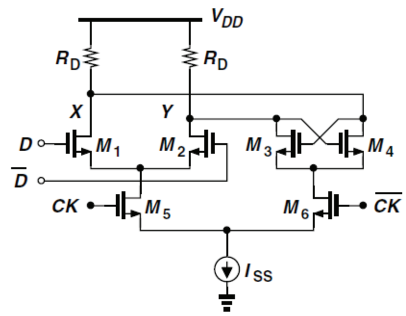
$$Y = AB$$

Divider Logic Style: CML (2)

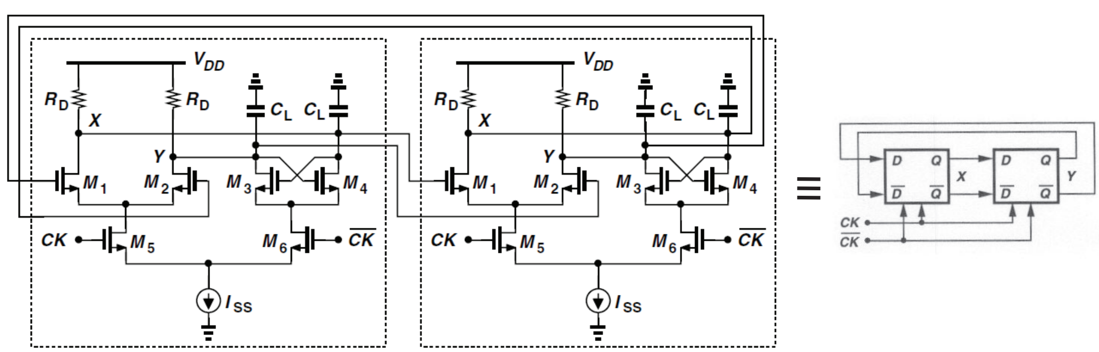
CML XOR Circuit



CML Latch



Divider Logic Style: CML Divide-by-2



• Typical Single-ended Swing: $R_D I_{SS}$ = 300mV

Design Procedure:

Given design constraint of power, swing, load cap

(1) Select I_{SS} with given power budget
(2) Select R_D to have $R_D I_{SS}$ ~ 300 mV
(3) Select $(W/L)_{1,2}$ for easy swing
(4) Select $(W/L)_{3,4}$ for sufficiently fast regenerative loop gain
(5) Select $(W/L)_{5,6}$ for the clocked pair to steer current easy and fast

Design Trade-offs:

• Higher I, Lower R → higher speed, higher current consumption
• Lower I, Higher R → lower speed, lower current consumption

PLL Design Example:
2.4GHz RF PLL Synthesizer

PLL Design Examples

- Application System
  – IEEE 802.15.1 (Bluetooth in 2.4GHz ISM Band)
- Design Procedure
  – Frequency Plan
  – Architecture Selection
  – Divider Design
  – Loop Filter Design
  – System Simulation

Frequency Plan and Lock Time

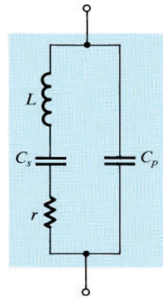
- Bluetooth Standard IEEE 802.15.1
  - 16 channels
  - fc=2405+5x(n-11), where n=11,12,….,26
    - fc=2405,2410, …. , 2480 MHz
  - Frequency hopping: 1600 hops/sec = 640 us dwell-time

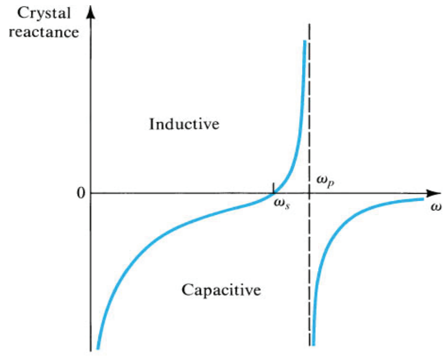
⇨ f_{REF} = 5MHz in order to support the required channel spacing

Reference Oscillator

- Typical Characteristics
  - High-Q Crystal (Q>10000)
  - Nominal operating frequency in the range of 1 ~ 100 MHz
- Types
  - Temperature Compensated Crystal Oscillator (TCXO)
  - Digitally-controlled TCXO (DTCXO)
  - Voltage-controlled TCXO (VCXO)
  - Oven-controlled TCXO (OCXO)
- Performance Parameters
  - Frequency tolerance over temperature, load, voltage
  - Frequency tuning range by control voltage
  - Phase noise
  - Supply voltage and current consumption
  - Form Factor (dimension)

Piezoelectric Crystal (Quartz)





$$Z(s) = \frac{1}{sC_p + \frac{1}{sL + 1/sC_s}}$$

Series resonance

$$\omega_s = \frac{1}{\sqrt{LC_s}}$$

Parallel resonance

$$\omega_p = \frac{1}{\sqrt{L(C_s//C_p)}}$$

since $C_p >> C_s$, $\omega_s \approx \omega_p$

- Crystal behaves inductive over a very narrow well-defined frequency range. → use it as a tank inductor for an oscillator.

Example: TCXO Datasheet

Temperature Compensated Crystal Oscillators (TCXO)
Surface Mount Type TCXO (LSI Type) KT2520Y Series
2.5×2.0mm

KYOCERA

Features
- Ultra-miniature SMD type (2.5×2.0×0.9mm)
- AFC function available
- Frequency stability : ±2.0×10⁻⁶/–30 to +85°C
- 2.3 to 3.5V drive available
- Reflow compatible

Applications
- 3G (CDMA, W-CDMA), GPRS, GSM

How to Order

KT2520Y ① 26000 ② D ③ C ④ W ⑤ 28 ⑥ T ⑦ AA ⑧

①Series
②Output Frequency
③Frequency Tolerance

B	$\pm1.0\times10^{-6}$
C	$\pm1.5\times10^{-6}$
D	$\pm2.0\times10^{-6}$

④Lower Operating Temp.

C	−30°C
E	−20°C
G	−10°C

⑤Upper Operating Temp.

W	+85°C
V	+80°C
U	+75°C

⑥Supply Voltage

28	2.8V	30	3.0V

⑦Voltage Control Range

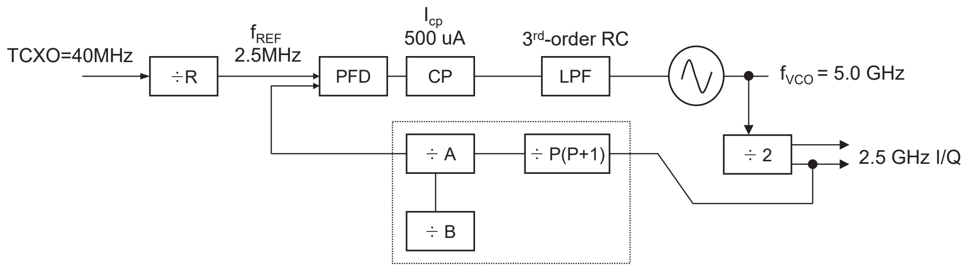
TCXO	T
VCTCXO	Customer Spec

⑧Option Code

Specifications

Item	Symbol	Conditions	Min.	Max.	Units
Output Frequency	Fo		12.6	40	MHz
Frequency Tolerance	F_tol	vs Temperature	−2	+2	$\times10^{-6}$
		vs Load	−0.2	+0.2	
		vs Voltage	−0.3	+0.3	
Frequency Aging	F_aging	Per Year	−1	+1	$\times10^{-6}$
Storage Temperature Range	T_stg		−40	+85	°C
Operating Temperature Range	T_use		−30	+85	°C
Voltage Control Range	Δf/V	Positive	±8	±15	$\times10^{-6}$
Supply Voltage	Vcc		2.3	3.5	V
Output Level	Vpp	10k ohm // 10pF	0.8	—	Vp-p
Current Consumption	I_{DD}		—	2	mA
Symmetry	SYM	@50% V_{DD}	40	60	%
Harmonics	—		—	−5	dBc

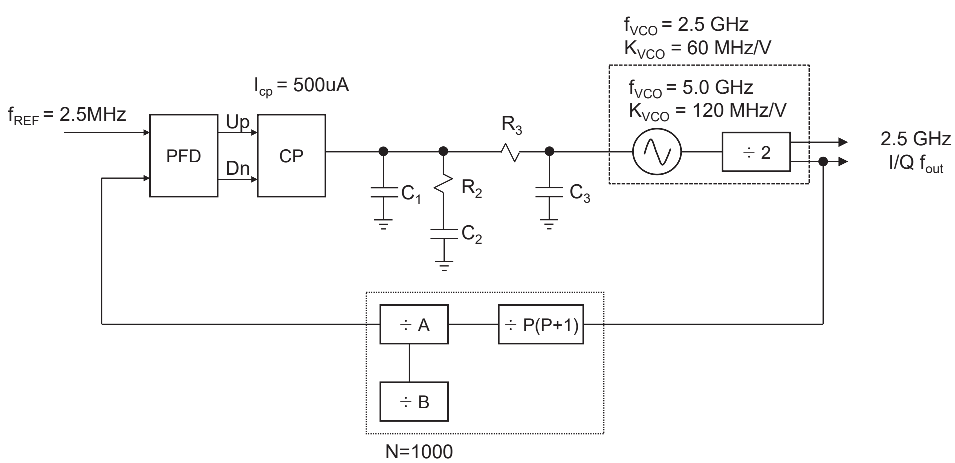
2.4-GHz PLL Architecture



Design Features:

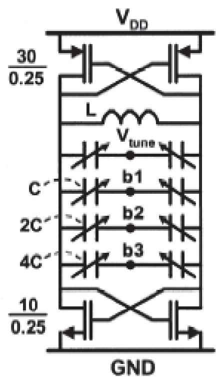
- f_{REF}=2.5MHz for the major reference spur on the channel boundary
- $f_{RF}=f_{VCO}/2$
  - VCO pulling mitigation
  - Easy I/Q generation
  - Kvco: 120 MHz at 5GHz, equivalently 60 MHz at 2.5GHz
- Integer-N Pulse-swallow type
- Charge-Pump Current Icp = 500 uA
- Passive Loop Filter: 3^{rd}-order RC LPF

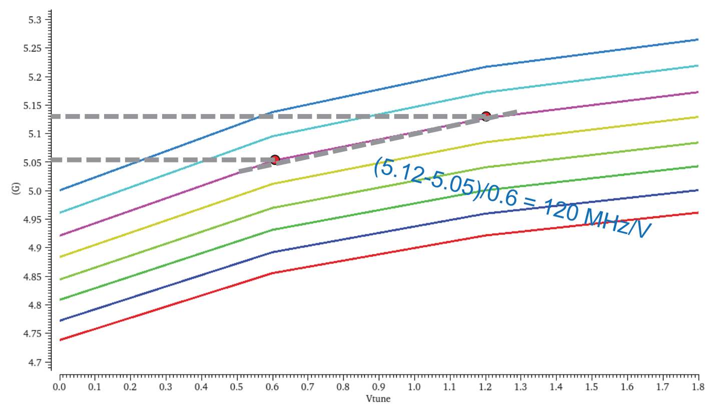
2.4GHz PLL Design Conditions



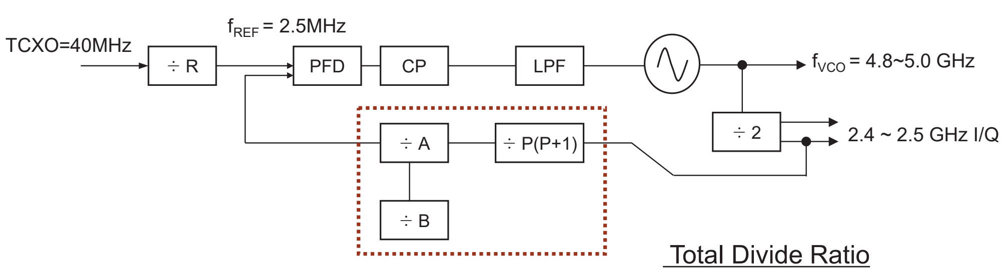
VCO Requirements

•Tuning range : 400MHz (= 4.8~5.2 GHz)
•K_{vco} = 120 MHz/V @ 5GHz (or, equivalently 60MHz/V @ 2.5GHz)
• Automatic self-tuning capability
• $\mathcal{L}$(4.9GHz) = -120dBc/Hz @ 1MHz offset
• $\mathcal{L}$(5.2GHz) = -118dBc/Hz @ 1MHz offset
⇨ Capacitor Bank is needed to cover the whole bands with the limited Kvco





Pulse-Swallow Counter & Feedback Divider Design



Total Divide Ratio
N = P(A-B)+(P+1)B = PA+B

Integer-N type Divider Design

Prescaler (P)	16/17
Programmable Counter (A-counter)	6 bit (60 ~ 63)
Swallow Counter (B-counter)	4 bit (1 ~ 16)
N = PA+B = 960~1000	960~1000

Loop Filter Design Issue

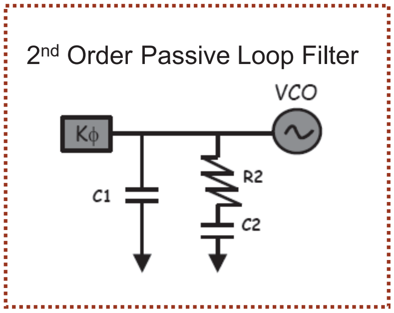
❖Loop filter is a key factor to determine the overall PLL transfer function.

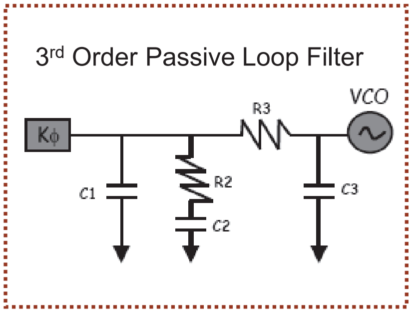
❖Loop filter design involves appropriate selections of

→ a proper filter topology, filter order, phase margin, loop bandwidth, and pole ratio.

❖ Fundamental Trade-off

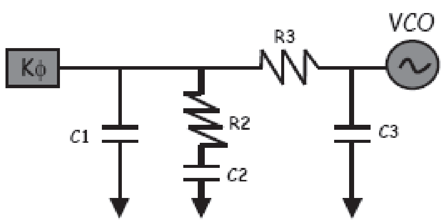
- Wide Loop Bandwidth → fast lock time but poor out-of-band noise/spur rejection
- Narrow Loop Bandwidth → slow lock time but good out-of-band noise/spur rejection





※Note that R_1C_1 and C_2 are relabeled by R_2C_2 and C_1, which is also often used in s ome textbooks as well as in MATLAB examples.

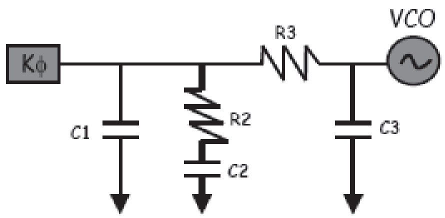
Loop Filter Design

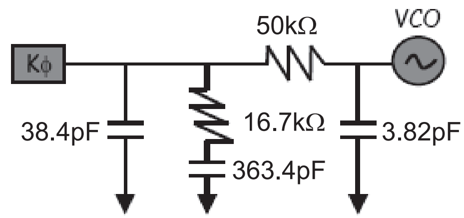


Loop Filter Design

Parameters	Considerations	Decision
Filter Topology	Active vs. Passive	Passive
Loop Filter Order	Spur reduction by R_3C_3	3^{rd} Order
Phase Margin	stability vs. lock-time (40~55°)	55°
Loop Bandwidth	lock-time vs. spur & noise	100kHz

3rd-Order Loop Filter Component Calculation





$$Z(s) = \frac{1+sT_2}{s(1+sT_1)(1+sT_3)} \cdot \frac{1}{C_{tot}}$$

$$K_{VCO} = K_V \times 2\pi, K_\Phi = I_{CP}/2\pi, \omega_P = 2\pi \times LBW$$

$$T_1 = \frac{\sec(\phi_M) - \tan(\phi_M)}{\omega_P}$$

$$T_3 = \sqrt{\frac{10^{\frac{ATTEN}{20}} - 1}{(2\pi f_{ref})^2}}$$

$$\omega_C = \frac{\tan(\phi_M) \times (T_1 + T_3)}{(T_1+T_3)^2 + T_1 T_3}\left[\sqrt{1 + \frac{(T_1+T_3)^2 + T_1 T_3}{[\tan(\phi_M)(T_1+T_3)]^2}} - 1\right]$$

$$T_2 = \frac{1}{\omega^2 (T_1 + T_3)}$$

Initial Value Determination

$$C_1 = \frac{T_1 K_\Phi K_{VCO}}{T_2 \omega_C^2 N}\left[\sqrt{\frac{1+\omega_C^2 T_2^2}{(1+\omega_C^2 T_1^2)\times(1+\omega_C^2 T_3^2)}}\right]$$

$$C_2 = C_1\left(\frac{T_2}{T_1} - 1\right)$$

$$R_2 = \frac{T_2}{C_2}$$

$$C_3 = \frac{T_3}{R_3} \text{ assuming to have } R_3 \approx R_2$$

Ref: D. Banerjee

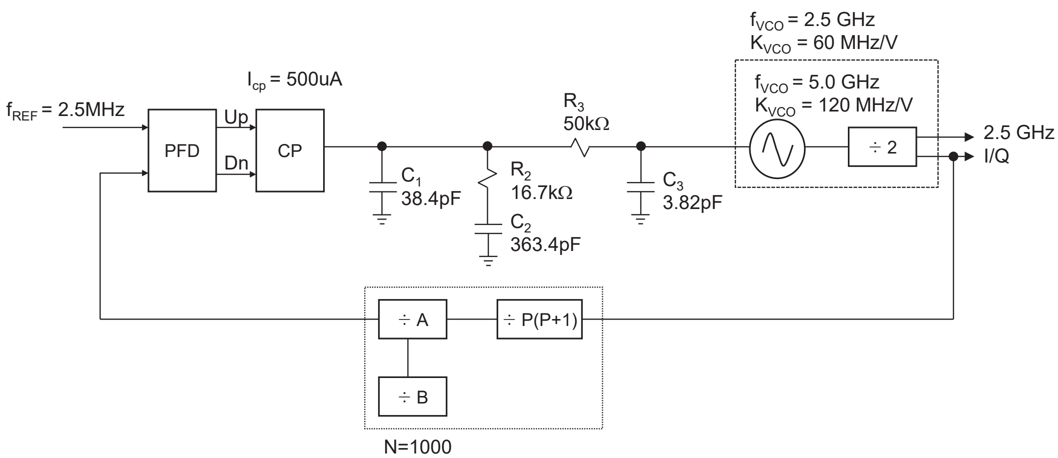
PLL System Design & Simulations

PLL System Simulation Before Circuit Simulation	
Simulation Tool	Matlab & Simulink
Loop Characteristics	Open and Closed Loop Phase Margin Loop Bandwidth Step Response for Lock Time Phase Noise
Behavioral Simulation	Lock Time

Key Loop Parameters

f_{REF}	2.5 MHz
fout	2.5 GHz
N	1000
Loop Bandwidth	100 kHz
Phase Margin	50°
Kvco	60 MHz/V
Icp	100 uA

2.4GHz PLL Design Results



Phase-Domain Linear Analysis

Matlab Example: Model PLLs in the Phase Domain

Model PLLs in the Phase Domain

R2024a

This example shows how to model a phase-locked loop (PLL) in the phase domain, compare the analytic results to simulation results in the time domain, and identify the advantages and disadvantages of each approach. Most PLL analysis is performed in the phase domain, and this example shows you how to produce a complete analysis using a minimum of time and effort.

The phase domain analysis calculates the PLL transfer function, loop error function, lock time and noise transfer impedances using the Control System Toolbox™. (Automated loop tuning is described in the Tune Phase-Locked Loop Using Loop-Shaping Design example.)

The PLL design and the time domain simulation model were derived from the Phase Noise at PLL Output example.

The general loop filter modeling procedure uses the Mixed-Signal Blockset™ Linear Circuit Wizard and was derived from the Circuit Design Details Affect PLL Performance example.

This example uses:
Mixed-Signal Blockset
Control System Toolbox

Open Script

Phase Domain Block Models

This section shows how to create the phase domain models for the individual blocks of an analog phase/frequency locked loop.

The term phase domain refers to the representation of a periodic signal in terms of its phase with respect to an ideal reference with the same period, instead of in terms of its voltage or current as a function of time. Most PLL analyses (in contrast with simulations) treat the PLL as a linear control system in the frequency domain, with the reference signal and the output of the VCO/divider represented in the phase domain.

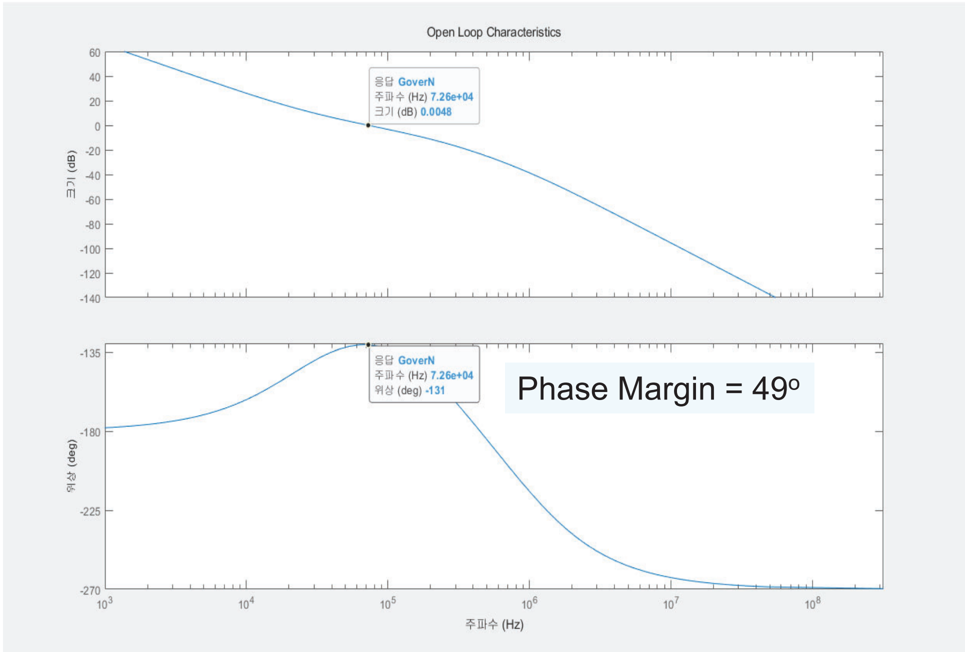
The Control System Toolbox supports both transfer function (tf) and state space (ss) models. While the tf objects tend to be easier to use, the ss objects are often more accurate at higher frequencies.

편집기 - C:₩Users₩USER₩Documents₩MATLAB₩hs_PLLMatlabSim₩hs_PLLMatlabSystemSim.m

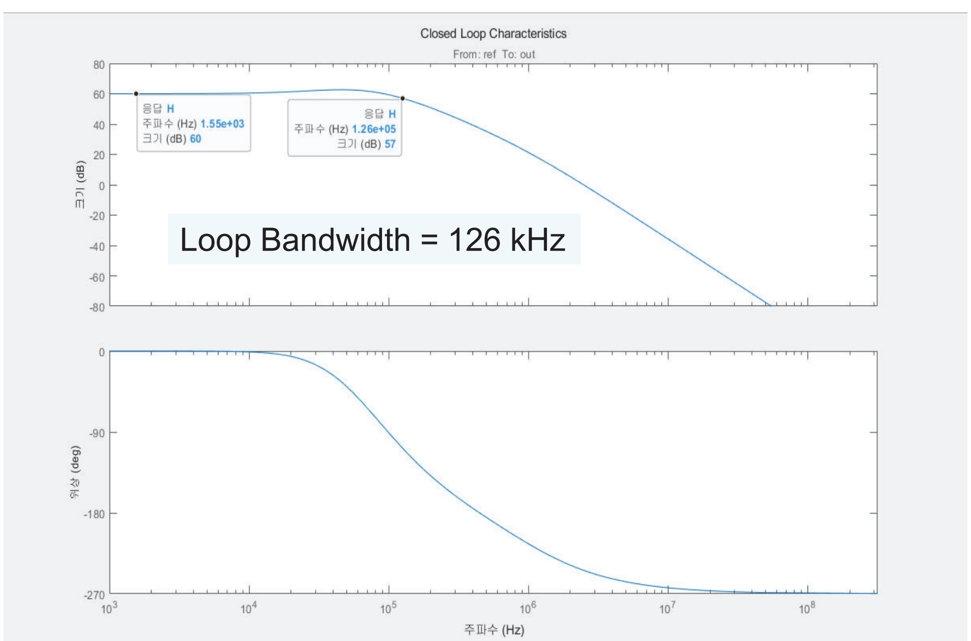
hs_PLLMatlabSystemSim.m

```
% <<../SimplePLLwithDivider.png>>
%%
% To enable comparison of analysis results to time domain simulation
% results, this example uses the same design values as the Phase noise at
% PLL output example.

%%
PllKphi = 0.5e-3;     % Charge Pump output current
PllKvco = 60e6;      % VCO sensitivity 100MHz/V @ 2.5GHz, 200MHz/V @ 5GHz
PllN    = 1000;       % Prescaler ratio

PllC1   = 38.4e-12; % Loop filter direct capacitance (F)

PllR2   = 16.7e3;   % Loop filter resistance for second order response (ohms)
PllC2   = 363.4e-12; % Loop filter capacitance for second order response (F)
PllR3   = 50e3;    % Loop filter resistance for third order response (ohms)
PllC3   = 3.82e-12; % Loop filter capacitance for third order response (F)

PllR4   = 0;        % Loop filter resistance for fourth order response (ohms)
PllC4   = 0;        % Loop filter capacitance for fourth order response (F)
%%
% *Reference*
%
% The reference signal is assumed to have the same frequency as the ideal
% reference used to define the phase domain. However, there can be a
% time-varying phase offset between the reference signal and the ideal
% reference. This phase offset $\theta_{ref}(t)$, or $\theta_{ref}(s)$ in
% the Laplace domain, is an input to the linear control system.
%
```

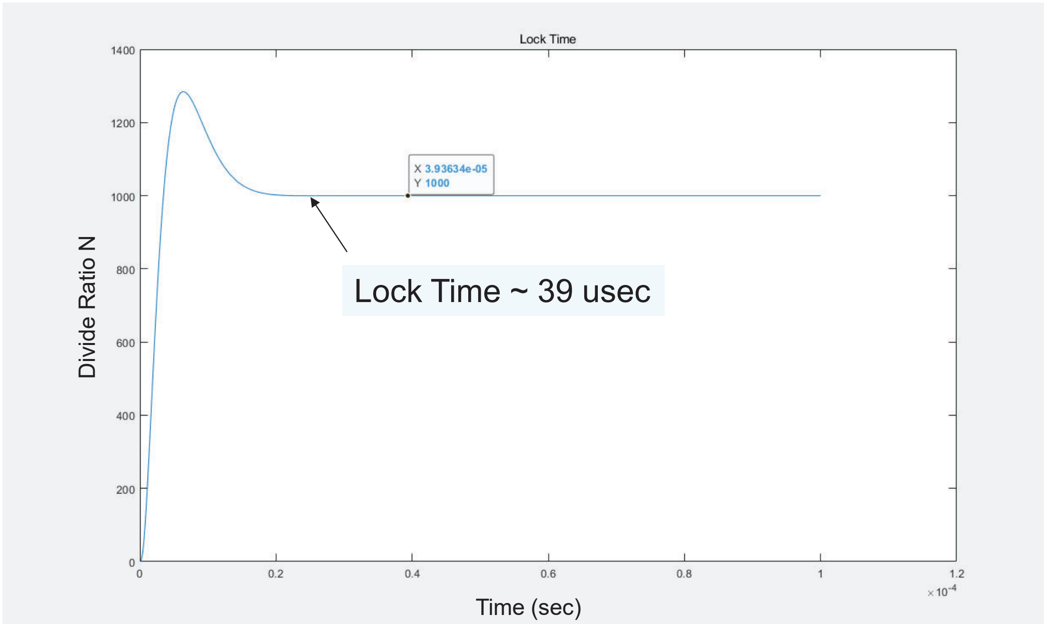
Open-Loop Gain



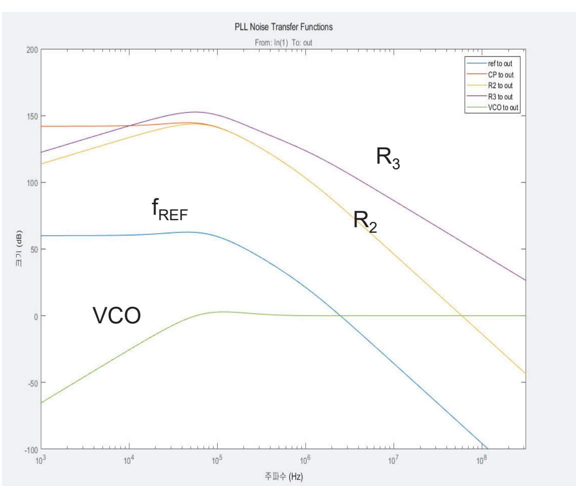
Closed-Loop Gain

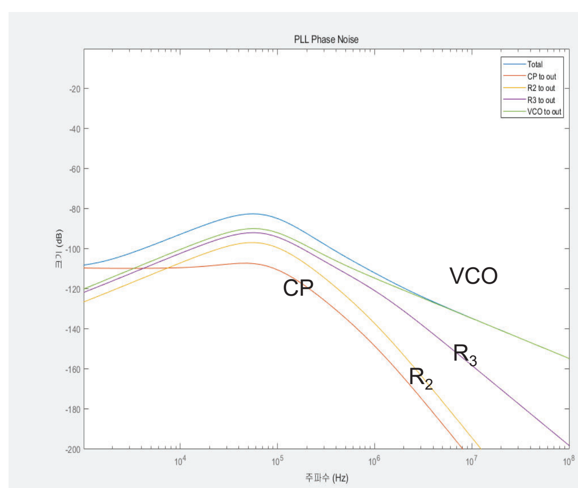


Lock Time by Unit Step Response

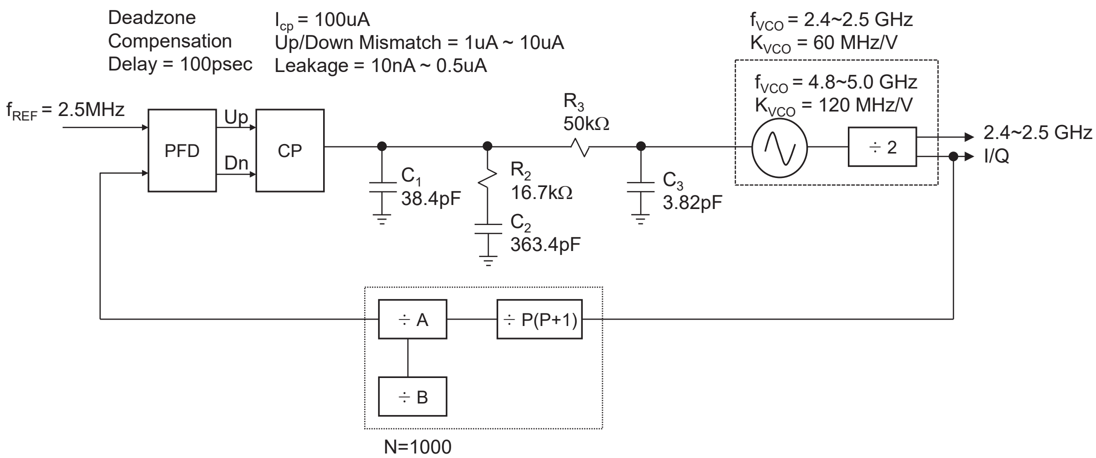


Phase Noise Transfer Characteristics

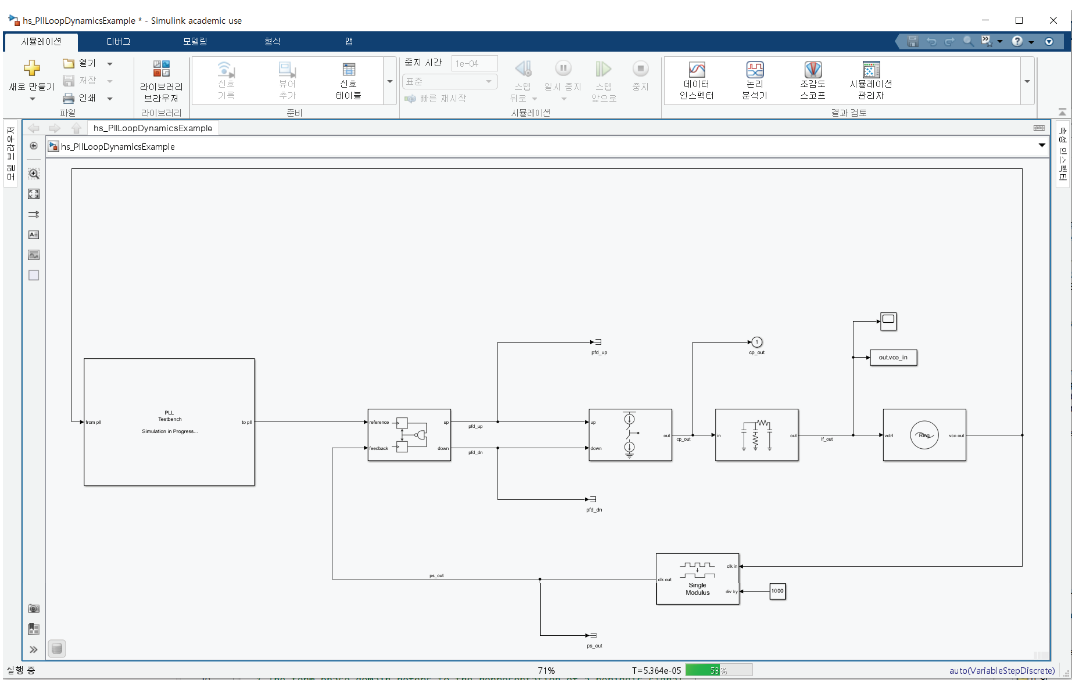




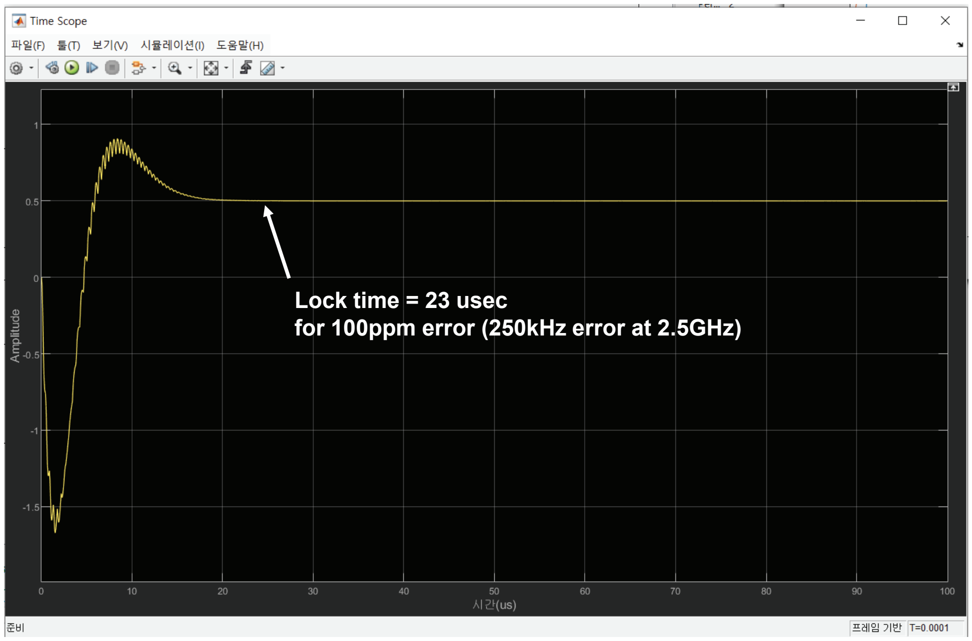
2.4GHz PLL with PFD/CP Non-idealities



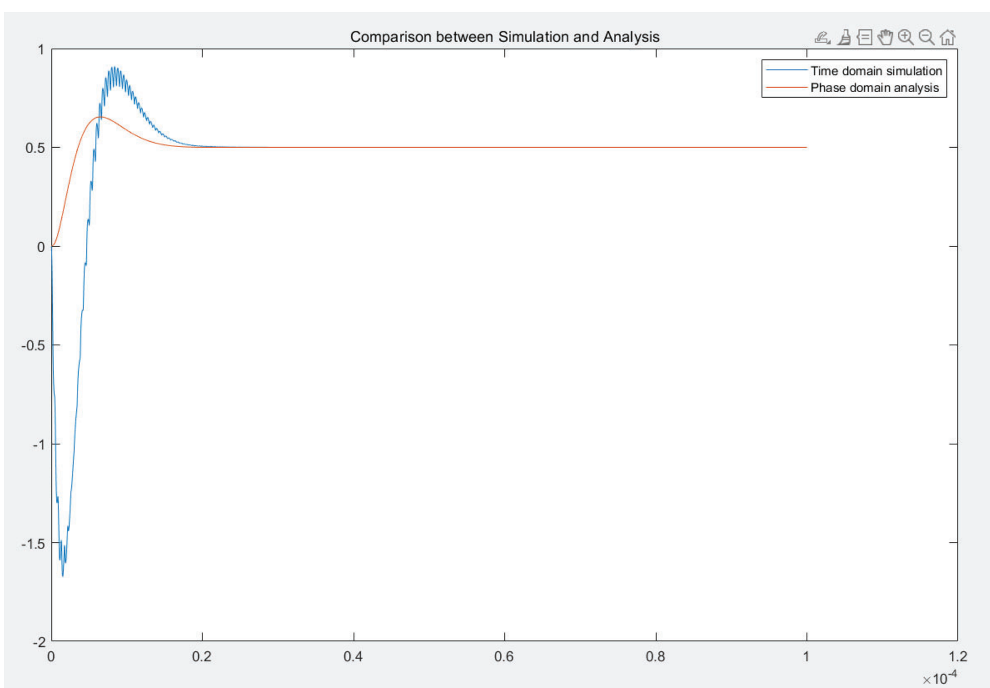
Simulink Time-Domain Analysis



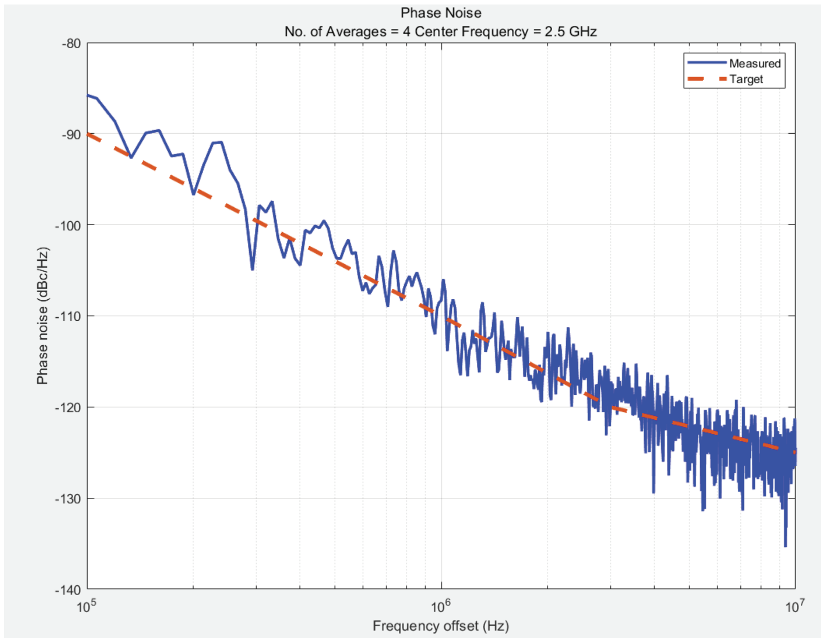
Lock Time



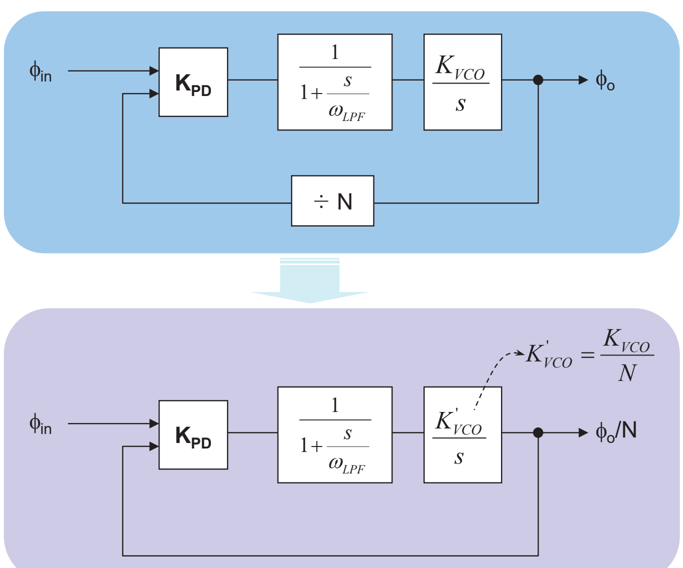
Locking Behavior Comparison



Phase Noise



Fast Closed-Loop PLL Time-Domain Top Simulation

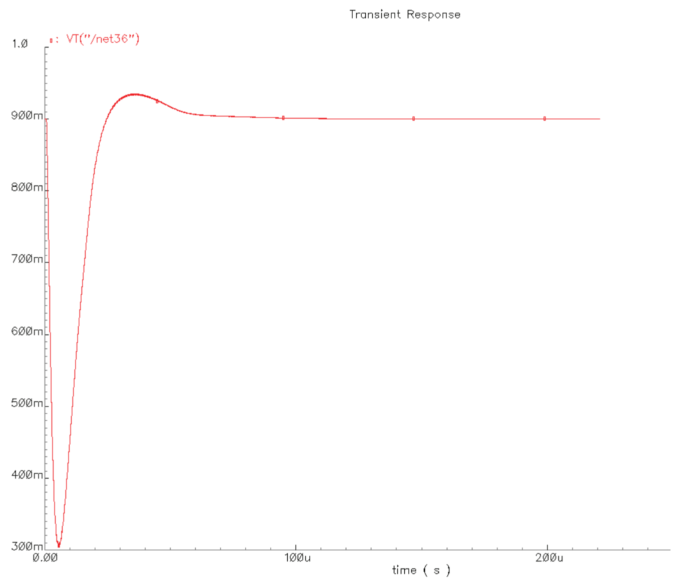


$$\frac{(\omega_{out}/N)}{\omega_{in}}(s) = \frac{K_{PD}K'_{VCO}}{\frac{s^2}{\omega_{LPF}} + s + K_{PD}K'_{VCO}}$$

- Dividers is removed by dividing the VCO gain by the divide-ratio
- VCO in AHDL Model

Fast Transient Simulation Result

- Locking simulation is done only in several minutes



Summary

- PLL-based frequency synthesizer is a key building block in a modern RF transceiver
- Frequency synthesizer performances directly affects the quality of communication
- Charge-pump PLL is the most widely used architecture.
- Integer-N PLL design involves trade-off among phase-noise, lock time, spur suppression, and loop stability
- Circuit design issues of the PLL building blocks are addressed

- References:

  - B. Razavi, *RF Microelectronics, 2nd Ed.,* Pearson, 2012
  - B. Razavi, *Design of Analog CMOS Integrated Circuits, Chapters 14 & 15;* NY, McGraw-Hill, 2001
  - D. Banerjee, *PLL Performance, Simulation, and Design*, 6th Ed, 2023 (also available at http://DeanBanerjeePLL.com)